MIL-T-27892F(ASG)
15 DECEMBER 1967

Superseding
MIL-T-27892E(USAF)
2 June 1967

MILITARY SPECIFICATION

TANK, STORAGE, LIQUID OXYGEN TMU-7A/E

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 This specification covers a skid-mounted, 2,000-gallon tank for storing liquid oxygen, designated TMU-7A/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-S-300	SHEET AND TAPE, REFLECTIVE, NONEXPOSED LENS, ADHESIVE BACKING
O-T-634	Trichloroethylene, Technical
HH-I-551	Insulation Block, Pipe Covering and Boards, Thermal (Cellular Glass)
UU-T-81	Tags, Shipping and Stock
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof, For Packaging

Military

MIL-V-173	Varnish, Moisture-And-Fungus-Resistant (For The Treatment of Communications, Electronic, And Associated Electrical Equipment)
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-R-3065	Rubber, Fabricated Parts
MIL-T-5021	Tests, Aircraft And Missile Welding Operators, Qualifications
MIL-I-6866	Inspection, Penetrant Method Of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-H-27301	Hose Assembly, Metal, Cryogenic Liquid, Transfer
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, With Dispenser

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S Military Property
MIL-STD-143	Specifications And Standards, Order of Precedence For The Selection Of

FSC 3655

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MIL-STD-808	Finishes, Protective And Codes For Finishing Schemes For Ground And Ground Support Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS33586	Metals, Definition Of Dissimilar
MS33656	Fitting End, Standard Dimensions For, Flared Tube Connection And Gasket Seal

Air Force-Navy Aeronautical

AN929	Cap Assembly, Pressure Seal, Flared Tube Fitting

DRAWINGS

Ordnance Corps

C8984433	Seat, LOX Coupling, 3-1/2 Inch
C8984434	Nut, LOX Coupling, 3-1/2 Inch
C8984435	Cone, LOX Coupling, 3-1/2 Inch
C8984990	LOX Cap, 3-1/2 Inch
C8984991	LOX Plug, 3-1/2 Inch
C8987830	Assembly, LOX Coupling, Male Half, All Sizes
C8987831	Assembly, LOX Coupling, Female Half, All Sizes
B8987832	Gasket, LOX And LN2, All Sizes
B8987839	Ring, Retaining, LOX And LN2 All Sizes
C8987856	Cone, LOX Coupling, 1 Inch
C8987857	Nut, LOX Coupling, 1 Inch
C8987870	LOX Plug, 1 Inch

Air Force

49B6564	Ring-Eyebolt, 25,000 LB Cargo Tiedown
59C6671	Hose, Transfer — Cryogenic Liquid, Assembly Of
66C1627	Fitting Assembly — Vacuum Disconnect

PUBLICATIONS

U. S. Air Force Specification Bulletin

518	Cargo, Aircraft Compartment Dimensional Data

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

ASME Boiler and Pressure Vessel Code

Section VIII Unfired Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The tank shall consist of the following major components:

Description	See Requirements
Liquid storage tank assembly	3.8.1
Pressure buildup system	3.8.2
Control housing assembly	3.8.3
Control panel	3.8.4
Skid base	3.8.5.

* 3.2.1 Component schematic. Figure 1 is a component schematic of the tank lines, valves, relief devices, and instruments. The piping is shown on both ends for illustration purposes only. The actual tank design has all components with exception of the outer shell relief device located on one end of the tank.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with varnish conforming to MIL-V-173. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Wherever practicable, lightweight metals shall be used in the construction of the tank. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to environmental conditions.

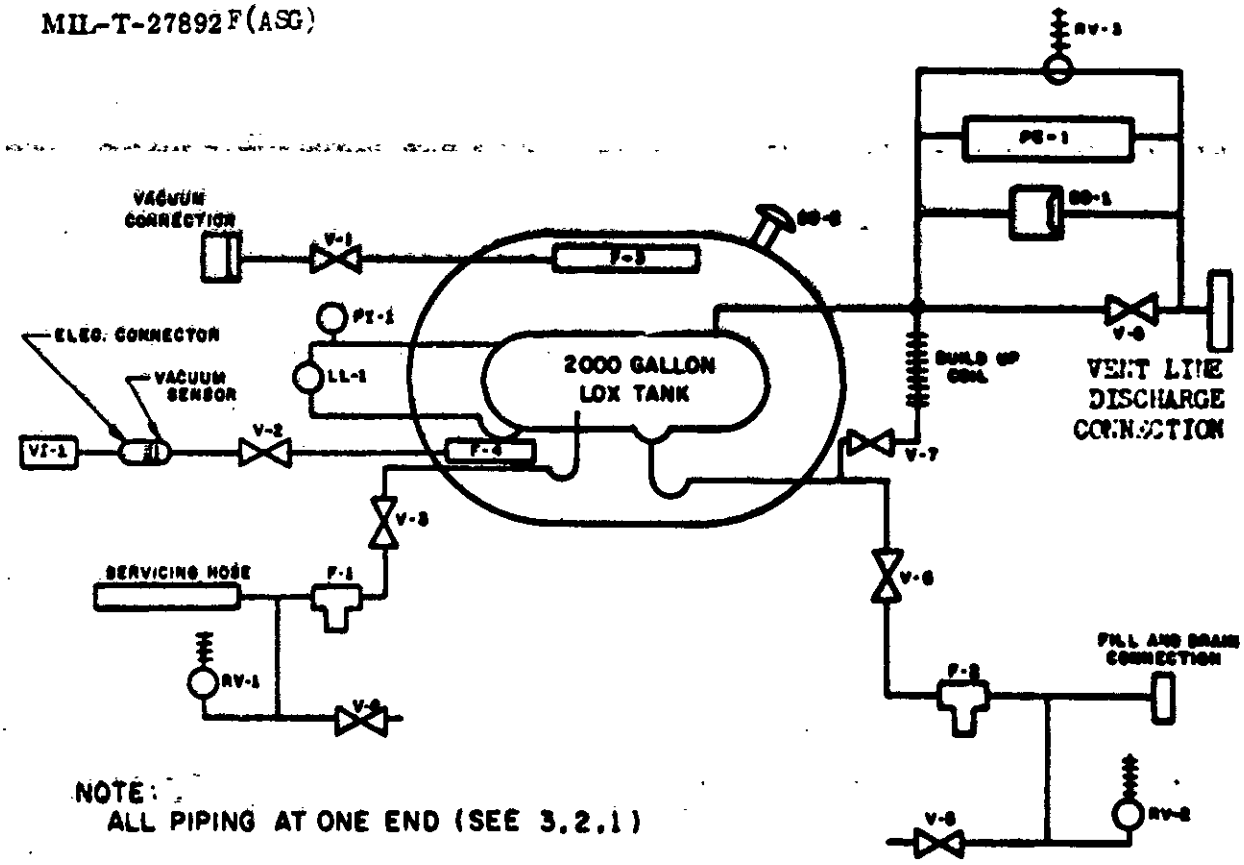
3.4.2.1 Chilled surfaces. All metallic surfaces of the tank and components thereof that contact high-purity oxygen or that are in a location where they will be chilled sufficiently during operation to cause condensation of moisture from the atmosphere shall be fabricated from corrosion-resistant materials that require no paint or other organic chemical coatings to protect them from corrosion.

3.4.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3 Gasketing and insulating materials. Plastic, rubber, or similar gasketing and insulating materials shall be compounded to insure their suitability for the intended application and, where applicable, their resistance to hydrocarbons or lower temperatures.

* 3.4.3.1 Nonmetallic components. Packings, gaskets, seals, and other nonmetallic components that come into contact with high-purity oxygen shall be compounded of materials compatible with use in a pure oxygen atmosphere under the conditions to which they will be subjected, and that are not adversely affected or deteriorated by continued use in an oxygen atmosphere. Polytetrafluoroethylene gasketing material for other than the B8987832 gasket shall be impregnated with not less than 25 percent pulverized glass fiber filler.

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SD-1 INNER SHELL SAFETY HEAD
SD-2 OUTER SHELL SAFETY HEAD
F-1 SERVICING LINE FILTER
F-2 FILL-DRAIN LINE FILTER
F-3 VACUUM FILTER (POWDER INSULATED TANK)
F-4 VACUUM INDICATOR FILTER (POWDER INSULATED TANKS)
LL-1 TANK LIQUID LEVEL INDICATOR
PI-1 VAPOR PHASE PRESSURE GAGE
RV-1 SERVICING LINE PRESSURE RELIEF VALVE
RV-2 FILL-DRAIN LINE PRESSURE RELIEF VALVE
RV-3 INNER SHELL PRESSURE RELIEF VALVE
V-1 VACUUM LINE SHUTOFF VALVE
V-2 VACUUM INDICATOR SHUTOFF VALVE
V-3 SERVICING LINE SHUTOFF VALVE
V-4 SERVICING LINE DRAIN VALVE
V-5 FILL-DRAIN LINE DRAIN VALVE
V-6 FILL-DRAIN LINE SHUTOFF VALVE
V-7 PRESSURE BUILDUP CONTROL VALVE
V-8 VAPOR VENT LINE SHUTOFF VALVE
VI-1 ANNULAR SPACE VACUUM INDICATOR
PC-1 ADJUSTABLE PRESSURE CONTROL

FIGURE 1. Component Schematic

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* 3.4.4 Rubber goods. All rubber goods such as door seals, et cetera, shall be fabricated to MIL-R-3065.

3.4.5 Combustible materials. Readily combustible materials shall not be used.

3.5 Design. The tank shall be a complete, self-contained liquid oxygen storage unit including all parts, controls, instruments, and accessories, designed for:

a. Being filled with liquid oxygen at a using site under high flow rates

b. Storing the liquid with low evaporation loss rates until needed, and at that time transferring the liquid into smaller servicing containers.

* 3.5.1 Reliability. The tank shall have a mean-time-between-failures (MTBF) of not less than 21 servicing cycles at 0.9 confidence. (See 6.3.1 and 6.3.2.)

3.5.2 Maintainability. Parts and assemblies shall be located and mounted to provide adequate clearance for repair and other maintenance and, where practicable, to permit removal and replacement of any part or assembly by removing or disconnecting only mounting bolts or fasteners, panel doors, tubing, or control cables to the part or assembly being removed.

3.5.2.1 Operating clearance. To the maximum practicable extent, maintainability provisions shall incorporate features insuring operating clearance for facilitating maintenance and servicing at low ambient temperatures (see 6.3.3) by personnel wearing heavy gloves or mittens and handicapped by bulky clothing and footgear.

3.5.2.2 Intricate devices. Intricate locking devices, controls, and threaded fastenings that can be easily overtorqued by personnel lacking feel through thick gloves or numbness shall be avoided where possible.

3.5.2.3 Covers and plate fasteners. Covers and plates that must be removed for component adjustments or for component or part removal shall be equipped with substantial quick-disconnect fasteners.

3.5.3 Detachable fittings. Wherever practicable, the tank shall be designed to include readily detachable- and attachable-type fittings in the liquid and vapor lines to permit rapid component removal and replacement. Disconnect points shall be clearly indicated and identified.

3.5.4 Protective covers. All cover plates, gaskets, and fittings necessary for protection of contained apparatus during operation, storage, and shipment of the tank shall be provided.

3.5.5 Common parts. The tank shall be designed to provide for the maximum practicable interchangeability of hardware and fastening devices by using the minimum number of types and sizes of bolts, screws, nuts, washers, and other common hardware.

3.5.6 Handling and transport provisions

3.5.6.1 Aircraft loading and transport. The tank design and configuration shall permit loading into and air transport by Types C-124 and C-133 aircraft.

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Applicable aircraft cargo compartment and loading door dimensional data is contained in USAF Specification Bulletin 518.

3.5.6.2 Transport acceleration forces. The tank shall be designed to withstand the acceleration forces specified in MIL-A-8421 when transported empty.

3.5.6.3 Hoisting and tiedown provisions. Tiedown provisions, including four or more tiedown ring assemblies conforming to Drawing 49B6564, shall be furnished as necessary for compliance with MIL-A-8421. Location of the tiedown ring shall prevent contact of the tiedown and tank shell. The tiedown provisions may serve as both the tiedown and hoisting provisions if the tank design and tiedown ring location will prevent damage when the tank is lifted by a single-hook overhead crane. The hoisting provisions shall permit lifting of the complete, assembled empty tank with a vertical acceleration of 2 gravitational units (g), and shall be so located on the tank that transportation personnel can easily rig safe slings from common cable and spreader bar components for hoisting by a single-hook overhead crane.

3.5.7 Lubricants. The tank shall be designed to fully comply with the requirements specified herein when components needing lubrication are serviced with military specification or federal specification lubricants in accordance with MIL-STD-838.

3.6 Construction. The tank shall be constructed to withstand the strains, shocks, vibrations, and other detrimental conditions incident to operation, maintenance, shipping, and storage with a minimum loss of service time for maintenance, repair, and periodic servicing.

3.6.1 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating position.

3.6.2 Component mounting. Components that need not be removed or repositioned for use shall be securely mounted to the liquid storage tank assembly or other major components in a manner that will insure against damage or unnecessary movement during operation and transport. All subbases and other fixtures needed for safe and secure component mounting shall be provided and installed.

3.6.3 Tubing and lines. Tubing and lines shall be located in protected positions, securely fastened to frame or body structures, and provided with metal protective loom or grommets at each point where they pass through members, except where a through-the-frame connector is provided.

3.6.4 Pressure vessels. Unless otherwise specified herein, the liquid storage tank assembly, piping, and connections thereto shall be fabricated in accordance with section VIII of the ASME Boiler and Pressure Vessel Code.

3.6.5 Controls, instruments, and accessories. All automatic and manual controls, instruments, fittings, safety devices, and accessories needed for safe and efficient operation of the tank shall be provided.

3.6.5.1 Installation. The controls, pressure gages, and instruments shall be installed in locations where they will not be contacted by liquid oxygen leaking from lines or connections and where water cannot drip on them as a result of vapor condensing on cold lines.

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3.6.6 Parts fabrication. Sheet-metal parts shall be of such thickness, rigidity, and strength as to withstand dents, warping, vibration, radiated heat, et cetera, encountered in service and maintenance and under the conditions specified herein.

3.6.7 Locking devices. Where practicable, all screws, pins, bolts, et cetera, shall be equipped with locking devices. Safety wire, self-locking nuts, cotter pins, lockwashers, et cetera, will be acceptable. Where practicable, lockwashers shall be secured to bolts or screws. Cotter pins shall be of a type fabricated from corrosion-resistant steel or other corrosion-resistant material.

* 3.6.8 Thread sealer. Tape conforming to MIL-T-27730 shall be applied to the threads prior to assembly of all pipe threaded fittings subject to contact by liquid or gaseous oxygen. The tape shall be applied starting with the third thread to prevent contaminating the system.

* 3.6.9 Cleaning solvent drains. Provisions shall be included to insure that cleaning solvent and flushing liquids can be readily and thoroughly removed from the liquid storage tank assembly, from the pressure buildup system, and from the associated tank piping and components following cleaning operations. Readily removable and replaceable threaded plugs or similar devices may be used in locations where it would be impractical to provide drain valves.

3.6.10 Pressure relief devices. Pressure relief devices shall be provided in any line or component that can be so isolated by closing valves, or otherwise, that dangerous pressures could develop. All pressure relief devices shall be installed at readily accessible locations. Provisions shall be incorporated to insure discharge of pressure without damage to equipment or danger to personnel. Safety valves shall be sized and selected in accordance with section VIII of the ASME Boiler and Pressure Vessel Code.

3.6.11 Certification of welders. All welding shall be accomplished by welders certified in accordance with MIL-T-5021, class A.

3.7 Performance. The tank shall meet the requirements specified herein when tested in accordance with the test methods specified under 4.6.

3.7.1 Storage capacity. The tank shall store under the conditions specified herein not less than 2,000 U. S. gallons of 99.5 percent pure liquid oxygen at its atmospheric pressure boiling temperature. The liquid storage tank assembly shall have an internal volume of not less than 2,100 U. S. gallons in order to provide an expansion (vapor) space of not less than 5 percent of the total design capacity.

3.7.2 Heat leak evaporation rate. The tank heat leak evaporation loss rate shall not exceed the equivalent of 70 pounds of oxygen per 24-hour day when:

(a) The liquid storage tank inner shell is filled to not less than 50 percent of design capacity with liquid oxygen or liquid nitrogen at atmospheric pressure

(b) The tank is in a location where the ambient temperature is not less than 125°F

(c) The upper surface of the tank is continuously exposed to solar or equivalent heat radiation of 100 to 120 watts per square foot for not less than 8 hours each day

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(d) Not less than 30 days have elapsed since evacuation of the tank annular space, and

(e) The tank has been subjected to all tests specified herein.

3.7.3 Vacuum retention. The tank annular space absolute pressure shall not rise above 100 microns Hg or show any increase attributable to leakage or outgassing within 7 days (168 hours) when:

(a) Prior to the 7-day test period, the tank annular space is evacuated to insure an absolute pressure of less than 100 microns of mercury (Hg) within 24 hours after the warm tank is filled with liquid oxygen or liquid nitrogen to 40 percent or more of design capacity

(b) The vacuum line is sealed after evacuation to prevent further evacuation prior to completion of the 7-day test period and the final absolute pressure measurement

(c) The vacuum retention testing specified under 4.6.6 is accomplished during the 7-day test period

(d) The tank inner shell is pressurized to not less than 30 pounds per square inch (psi) with oxygen or nitrogen gas throughout the 7-day test period, and

(e) The tank is subjected to ambient temperatures throughout the test period.

3.7.4 Air transportability. The tank shall not be damaged by the air transport acceleration forces specified in MIL-A-8421 when transported empty.

3.7.4.1 The empty tank shall also withstand, without damage, like forces as imposed by the flight and taxiing loads specified in MIL-A-8421 while being transported on a railroad car, seagoing vessel, or truck.

3.8 Details of components

3.8.1 Liquid storage tank assembly

3.8.1.1 Liquid storage tank assembly components. The liquid storage tank assembly shall consist of the following:

(a) Inner shell

(b) Outer shell

(c) Inner shell suspension system

(d) Insulation

(e) Manway

(f) Piping and connections

(g) Safety devices

(h) Controls

(i) Instruments.

3.8.1.2 Fabrication. The liquid storage tank assembly shall be a welded vessel designed and fabricated in accordance with section VIII of the ASME Boiler and Pressure Vessel Code to contain, and retard evaporation loss of, liquid oxygen as necessary to insure compliance with the performance requirements specified in 3.7.1 and 3.7.2. In addition, the inner shell shall be certified under the code for the specified working pressure.

3.8.1.2.1 Component arrangement. The liquid storage tank assembly shall be fabricated with the inner shell suspended within the outer shell to maintain a vacuum tight annular insulation space of not less than 8 inches between corresponding points on the shells and heads.

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*3.8.1.2.2 Dissimilar metal vacuum joints. Any dissimilar metal vacuum joints (see 6.3.4) used in the tank construction shall withstand the following without developing leakage detectable with a helium mass spectrometer operated at its maximum sensitivity:

(a) Thermal shocks from 125°F to the atmospheric pressure boiling temperature of liquid nitrogen

(b) Axial vibration at a frequency of 60 cycles per second (cps) and a double amplitude of not less than 0.050 inch with only one end of the joint fastened to the vibration input device and while submerged in liquid nitrogen at its atmospheric pressure boiling temperature

(c) Lateral vibration at a frequency of 60 cps and a double amplitude of not less than 0.200 inch with only one end of the joint fastened to the vibration input device and while submerged in liquid nitrogen at its atmospheric pressure boiling temperature

3.8.1.3 Inner shell. The inner shell shall physically store the liquid oxygen. Insofar as practicable, it shall be designed to have the minimum surface area consistent with the requirements specified herein.

3.8.1.3.1 Attachments. The inner shell shall be equipped with all parts, fittings, and accessories necessary to provide for support, entry for cleaning, liquid transfer, total draining, venting, safety, and determination of liquid level. The inner shell shall include sump and drain provisions.

3.8.1.3.2 Inner shell material. The inner shell shall be fabricated from material with sufficient strength, rigidity, and vacuum-holding properties to insure compliance with the requirements specified herein, and that is resistant to corrosion caused by contact with liquid or gaseous oxygen or nitrogen, moist air, water, or cleaning solvents such as trichloroethylene.

3.8.1.3.3 Inner shell design pressure (see 6.3.5). The inner shell shall be designed and constructed for a maximum working pressure adequate to permit the vapor space pressure relief valve setting specified in 3.8.1.9.3.1.1 and for a normal working pressure of not less than 50 pounds per square inch gage (psig) for transferring liquid oxygen under the conditions specified herein. Design and construction shall also be such as to insure against:

(a) Distortion or damage when the liquid storage tank assembly annular insulation space is evacuated to a high vacuum or is pressurized to the outer shell pressure relief device maximum opening pressure, and

(b) Collapse of or damage to the inner shell when it is evacuated to an absolute pressure of 100 microns Hg or less with the annular insulation space at atmospheric pressure.

3.8.1.3.4 Sump and drain provisions. The inner shell shall be constructed with a sump, to which the liquid fill-drain line shall connect, and a smooth inside bottom surface throughout an arc of not less than 6 inches on each side of the vertical centerline for the entire length of the inner shell to insure complete drainage into the sump. The sump shall be not less than 7-1/2 inches in diameter.

3.8.1.4 Outer shell. The outer shell shall provide support for the inner shell

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and inner shell suspension system. The underside of the outer shell shall be reinforced as necessary to comply with the requirements specified herein, and shall be equipped with all fittings and attachments needed for mounting the tank on a flat concrete pad or other suitable surface.

3.8.1.4.1 Outer shell material. The outer shell shall be fabricated from material having sufficient strength, rigidity, and vacuum-holding properties to insure compliance with the requirements specified herein.

3.8.1.4.1.1 The outer shell sections through which piping connections to the inner shell pass shall be fabricated from material resistant to corrosion caused by contact with oxygen, moist air, salt atmosphere, or water, to protect against corrosion in case the protective coating is damaged by the extreme changes in temperature that it must undergo on and adjacent to the lines carrying liquid and cold vapors. The use of corrosion-resistant, low thermal conductive sleeves to thermally isolate the liquid- and vapor-carrying lines from the outer shell will be acceptable.

3.8.1.4.2 Outer shell design pressure. The outer shell shall be designed and constructed for a working pressure that will insure sufficient strength and rigidity to prevent distortion or damage when the annular insulation space is evacuated to specified vacuum or pressurized to the outer shell pressure relief device maximum opening pressure.

3.8.1.4.3 Vacuum line. The outer shell shall be provided with a vacuum line and fittings designed for effective evacuation of the annular insulation space. The vacuum line shall incorporate a filter, a shutoff valve, and a vacuum line disconnect fitting. The vacuum line shall be installed to exhaust gases from the highest point of the annular insulation space, adjacent to the outer shell, and shall, in order to facilitate pumping down, incorporate features that will aid the flow of gases through the insulation to the vacuum line. All permanent joints in the vacuum line shall be welded. Disconnect points shall use flanged and bolted fittings.

3.8.1.4.3.1 Vacuum line shutoff valve. The vacuum line shutoff valve shall be of a type suited to high vacuum service, and that can be readily opened or closed with one hand. The valve shall seal the vacuum line as necessary for compliance with 3.7.3 when closed with one hand, and shall provide the least practicable resistance to gas flow when open.

3.8.1.4.3.1.1 Vacuum line shutoff valve mounting. The vacuum line shutoff valve shall be solidly mounted in an accessible location where it cannot be mistaken for a tank control valve. It shall be so mounted to supporting members that force applied in opening or closing the valve will not damage the vacuum line piping.

3.8.1.4.3.2 Vacuum line filter. The vacuum line filter shall be a removable flanged and bolted type unit extending into the tank annular space, and shall have an effective filtering area of not less than 500 square inches. The filter shall present the minimum practicable restriction to gas flow consistent with effectively preventing insulation passing from the annular space into the vacuum pump line. In addition, the filter shall be designed to withstand backflow of gases as would be associated with inadvertent opening of the vacuum shutoff valve to the atmosphere.

3.8.1.4.3.3 Vacuum line disconnect fitting. The vacuum line shall terminate in a fitting conforming to Drawing 66C1627.

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3.8.1.4.3.3.1 Vacuum line disconnect fitting mounting. The vacuum line disconnect fitting shall be mounted in a location readily accessible from outside the control housing assembly for attachment of a vacuum pump hose. The disconnect fitting shall be so mounted that forces applied to the fitting during attachment or removal of the vacuum pump hose, or while the vacuum pump hose is attached, will not damage the vacuum line.

3.8.1.5 Inner shell suspension system. The inner shell shall be supported and positioned within the outer shell by a suspension system that will thermally isolate the inner shell from the outer shell and provide an annular insulation space as specified in 3.8.1.2.1.

3.8.1.5.1 Strength. The inner shell suspension system shall be of sufficient strength and rigidity to solidly support and prevent damage to or unnecessary movement of the inner shell when it is filled with liquid oxygen to any level, or subjected to the forces specified in 3.7.4.

3.8.1.5.2 Heat transfer resistance. The inner shell suspension system shall be fabricated from material possessing the maximum practical resistance to heat flow, and shall provide the minimum practicable heat transfer area between the liquid storage tank assembly inner and outer shells without sacrifice of necessary strength and rigidity.

3.8.1.6 Annular space insulation. The liquid storage tank assembly annular insulation space shall be filled with an insulation that will, together with evacuation of the annular space, insure compliance with the requirements specified in 3.7.2. The insulating material shall be of a composition that will not spark or burn when touched with a glowing platinum wire in an atmosphere of pure oxygen. Application of the insulation shall be such as to insure against settling or displacement in service.

3.8.1.7 Manway. A manway with a free passage area of not less than 19-1/2 inches in diameter shall be installed in the top of the liquid storage tank assembly to permit entry for cleaning and inspecting the complete interior of the inner shell without breaking the annular insulation space vacuum. The manway joints, seams, and attachment points shall be smoothed and rounded after assembly as necessary to eliminate all sharp edges, corners, points, and other obstructions to safe entry. Use of the manway as an exit for lines leading from the inner shell will not be permitted.

3.8.1.7.1 Manway installation. All manway seams and all joints between the manway assembly and the inner and outer shells that will be exposed to annular insulation space vacuum shall be of welded construction. Gasketed or other nonpermanent-type seams, joints, or attachments will not be acceptable for sealing the annular insulation space vacuum.

3.8.1.7.2 Covers. Sturdy covers shall be provided as necessary to seal the manway against damage, liquid entry, and inner shell pressure loss during use of the tank under the conditions specified herein. If the cover arrangement seals the manway at the inner shell entrance in addition to sealing the manway at the outer shell entrance, the outer shell cover shall be equipped with a bursting disc or other positive sealing

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pressure relief device capable of preventing a pressure buildup between the manway covers that could damage the bellows specified in 3.8.1.7.3.

3.8.1.7.3 Bellows. A flexible bellows of sufficient diameter to permit ready installation and removal of the inner shell cover, if used, through the outer shell access hole shall be provided to seal the manway from the annular insulation space. A shield shall be fabricated inside the bellows to protect the bellows from damage during passage or manway cover removal.

3.8.1.7.4 Insulation. The space within the flexible bellows and between the manway covers or between the outer shell manway cover and the inner shell manway entrance covering device shall be packed or otherwise filled with a readily removable and replaceable insulation that will insure the minimum convection and conduction losses.

3.8.1.7.5 Lockout device. The manway cover installation provisions shall include a readily attachable lockout device, with duplicate keys, for use whenever a man is within the tank. The attached lockout device shall prevent installation of manway covers until unlocked by key and removed. Provisions shall be made for safe and secure storage of the lockout device when not in use.

3.8.1.8 Piping and connections

3.8.1.8.1 Intertank piping connections. Piping and connections between the inner and outer shells shall be held to the minimum necessary for insuring safety and performance in accordance with the requirements specified herein. Connecting piping shall be of corrosion-resistant material possessing the maximum practicable resistance to heat flow, and shall be installed to provide the maximum practicable length of heat transfer path combined with the minimum practicable heat transfer area.

3.8.1.8.1.1 Liquid lines. Separate liquid fill-drain and liquid servicing lines, valves, filters, and disconnects shall be provided. The centerline distance between the fill-drain line termination and the servicing line termination shall be a minimum of 9 inches.

3.8.1.8.1.1.1 Liquid fill-drain line. The liquid fill-drain line shall be sized and installed to provide for liquid oxygen flow or transfer into or out of the liquid storage tank assembly inner shell at a rate of not less than 100 gallons per minute (gpm) without the pressure drop through the line and filter exceeding 5 psig. The line shall terminate outside the liquid storage tank assembly outer shell in a 2-inch external pipe thread, and shall have a diameter of not less than 2 inches.

3.8.1.8.1.1.1.1 Liquid fill-drain line installation. The liquid fill-drain line shall be installed to drain liquid from the bottom of the sump specified in 3.8.1.3.4. In addition, the liquid fill-drain line installation shall be such as to:

(a) Insure against liquid entering the line from the inner shell when all shutoff valves are closed

(b) Insure that the inner shell and sump can be gravity drained to not less than 1/2 inch below the level of the inner shell inside bottom surface through the fill-drain line without pressurization when the tank is set on a level surface

(c) Minimize heat transfer between the inner and outer shells, and

(d) Expose the least practicable length of line outside the outer shell.

*(e) The liquid servicing line, valve, filter, and fittings outside the tank assembly outer shell shall be insulated with type II, class 1, pipe covering conforming to HH-I-551. The insulation shall be so applied, covered, and sealed that entrance of moisture will be prevented.

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3.8.1.8.1.2 Liquid fill-drain line shutoff valve. A manually operated shutoff valve having a free passage size approximately equal to that of the liquid fill-drain line inner diameter when fully open shall be provided in the liquid fill-drain line as near as practicable to the point where it enters the outer shell. The valve shall be so located and installed that liquid flow to the pressure buildup coil will not be restricted. General design and construction shall be as specified in 3.8.1.10.1 through 3.8.1.10.2.

3.8.1.8.1.2.1 Liquid fill-drain line shutoff valve mounting. The liquid fill-drain line shutoff valve shall be solidly mounted in a location readily accessible to an operator standing on the ground outside the control housing assembly. The liquid fill-drain line shutoff valve shall be so mounted to supporting members that force applied in opening or closing the valve will not damage the liquid fill-drain line piping.

3.8.1.8.1.3 Liquid fill-drain line disconnect coupling. A 3-1/2-inch, male, liquid-oxygen coupling half assembly consisting of a coupling seat, gasket, and cap conforming to Drawings C8984433, B8987832, and C8984990, respectively, assembled as shown on Drawing C8987830, shall be installed on the liquid fill-drain line external pipe thread using a bushing or similar device to adapt the 3-1/2-inch internal pipe thread of the coupling to the 2-inch external pipe thread on the liquid fill-drain line.

* 3.8.1.8.1.3.1 Liquid fill-drain line disconnect coupling mounting. The disconnect coupling shall be solidly mounted as near as practicable to the fill-drain valve in a location readily accessible to an operator standing on the ground outside the control housing assembly. The disconnect coupling mounting location shall provide sufficient clearance around the coupling for use of strap wrenches or other tools to assist in attachment or removal. The disconnect coupling shall be so mounted to supporting structures or members that a torque of 300 pound-feet (lb ft) per inch of nominal line size applied to the coupling in either direction of rotation with a 24-inch wrench will not damage tank piping or connections. The mounting of the fill drain line shutoff valve and of the disconnect coupling to supporting members shall be accomplished with material possessing considerable resistance to heat flow to reduce to a minimum evaporation of liquid filled into the tank through the liquid fill-drain line.

3.8.1.8.1.4 Fill-drain line filter. The fill-drain line shall be provided with a filter between the shutoff valve and the fill-drain line male coupling through which all liquid oxygen entering or leaving the tank through the fill-drain line disconnect coupling shall flow. The filter shall be located and installed to insure ready access for inspection, cleaning, and removal.

* 3.8.1.8.1.4.1 Liquid fill-drain line filter design. The filter shall be of either T or Y configuration, with the element constructed from stainless-steel or monel. Sintered or powdered metal material shall not be utilized unless backed up by a wire-wound or wire-mesh cloth element in which the wires have been fused under controlled conditions to positively prevent migration of particles that may be shed by the sintered or powdered metal portion. The filter element shall be fused to the housing in a manner which will prevent bypass of the element, as shown on figure 2. Elastomeric seals shall not be used in the filter design or construction. The proper direction of flow shall

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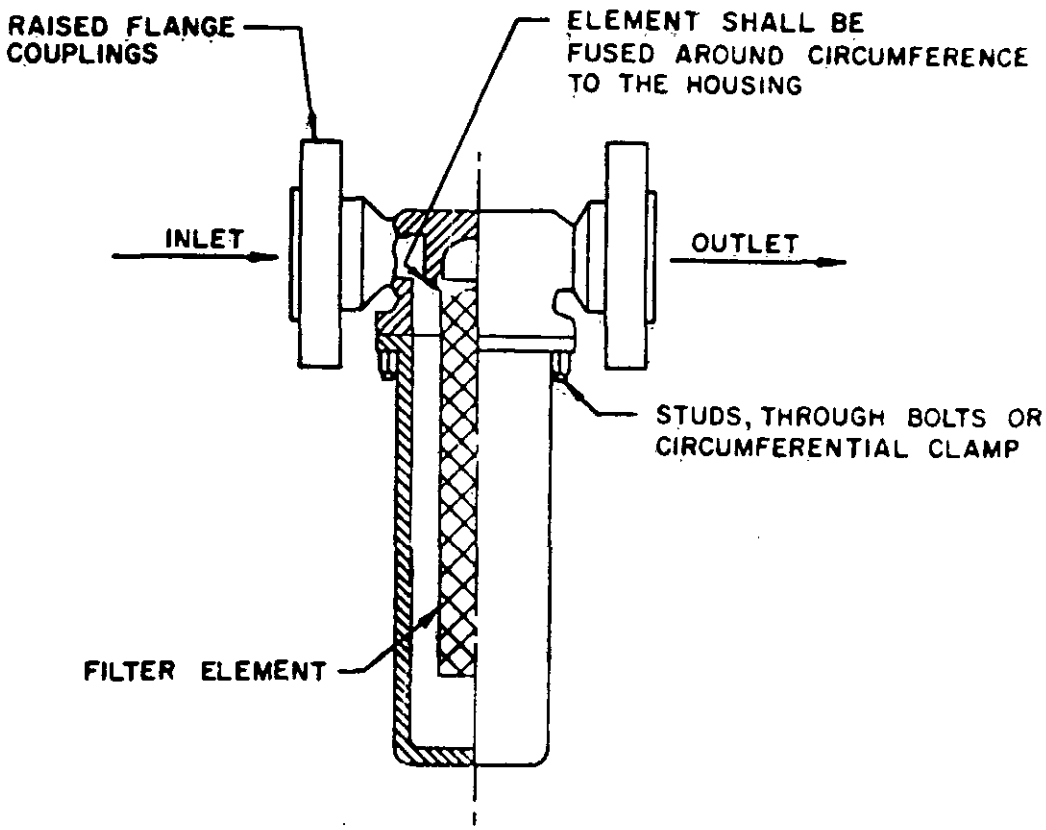


FIGURE 2. Filter Representative General Design and Construction

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be plainly marked on each side of the filter housing in a prominent location.

3.8.1.8.1.4.2 Liquid fill-drain line filter performance. The liquid fill-drain line filter shall remove 98 percent by weight of all particles whose smallest dimension is 10 microns or greater (10 micron nominal rating). The filter shall further remove all particles whose smallest dimension is 40 microns or greater (40 micron absolute rating). In addition, the filter shall be sized to pass liquid oxygen at a flow rate of 100 gpm without the pressure drop across the filter exceeding 3 psi when the filter is contaminated with 5 grams of particles having the size distribution specified in table I. The filter shall not permit bypass of the element under any combination of conditions, and shall be capable of withstanding a pressure drop of 50 psi across the element without damage.

TABLE I. Particle Size

Size of Particles (Microns)	Percentage by Weight
10 to 20	36 ±3
20 to 40	24 ±3
40 to 60	16 ±3
Over 60	24 ±3

3.8.1.8.1.5 Liquid servicing line. The liquid servicing line shall be sized and installed to provide for filling smaller tanks and portable liquid servicing trailers at rates up to and including 25 gpm without the pressure drop across the line and filter exceeding 5 psig. The liquid servicing line shall be not less than 1-inch diameter.

3.8.1.8.1.5.1 Liquid servicing line installation. The liquid servicing line shall be installed with a standpipe or other arrangement that will withdraw liquid at a height of 2 ±1/4 inch above the bottom of the inner shell to preclude withdrawing sediment with the liquid. In addition, the liquid servicing line installation shall be such as to:

(a) Insure against liquid oxygen entering the line from the tank inner shell when all liquid shutoff valves are closed

(b) Insure that the inner shell can be gravity drained to the level of the standpipe through the liquid servicing line without pressurization

(c) Minimize heat transfer between the inner and outer shells, and

(d) Expose the least practicable length of line outside the tank outer shell.

3.8.1.8.1.5.2 Liquid servicing line shutoff valve. A manually operated shutoff valve having a free passage size approximately equal to that of the liquid servicing line inner diameter when fully open shall be provided in the liquid servicing line as near as practicable to the point where it enters the outer shell. The valve shall be so located and installed that liquid flow to the pressure buildup coil will not be restricted. General design, construction, and mounting shall be as specified in 3.8.1.10.1 through 3.8.1.10.2.

3.8.1.8.1.5.2.1 Liquid servicing line shutoff valve mounting. The liquid servicing line shutoff valve shall be solidly mounted in a location readily accessible to an operator standing on the ground outside the control housing assembly. The liquid servicing line shutoff valve shall be so mounted to supporting members that force applied in opening or closing the valve will not damage the liquid servicing line piping.

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3.8.1.8.1.5.3 Servicing line disconnect fitting. The liquid servicing line shall terminate outside the liquid storage tank assembly outer shell in a 1-inch external pipe thread.

3.8.1.8.1.5.3.1 Servicing line disconnect fitting mounting. The servicing line disconnect fitting shall be solidly mounted as follows:

(a) As near as practicable to the servicing line shutoff valve in a location readily accessible to an operator standing on the ground outside the control housing assembly.

(b) Where the disconnect fitting will not be subjected to bending or other damaging stresses when the attached hose is secured in its storage provisions

(c) Where there will be sufficient clearance around the fitting for use of strap wrenches or other tools to assist in attachment or removal of the liquid servicing hose or the disconnect fitting itself, and

(d) Where sufficient clearance for attachment or removal of fill-drain hoses without interference will be insured.

The disconnect fitting shall be so mounted to supporting structures or members that a torque of 300 lb ft per inch of nominal line size applied to the fitting in either direction of rotation with a 24-inch wrench will not damage tank piping or connections. The mounting of the servicing line valve and of the disconnect fitting to supporting members shall be accomplished with material possessing considerable resistance to heat flow to reduce to a minimum evaporation of liquid serviced through the line and hose.

3.8.1.8.1.5.4 Servicing line filter. The servicing line shall be provided with a filter between the shutoff valve and the servicing line disconnect fitting through which all liquid oxygen transferred through the servicing line shall flow. The filter shall be located and installed to insure ready access for inspection, cleaning, and removal. The design of the filter shall be in accordance with 3.8.1.8.1.4.1.

3.8.1.8.1.5.4.1 Servicing line filter performance. The servicing line filter performance shall be as specified in 3.8.1.8.1.4.2, except that it shall be sized to pass liquid oxygen at a flow rate of 25 gpm without the pressure drop exceeding 3 psi when the filter is contaminated with 5 grams of particles having the size distribution specified in table I.

3.8.1.8.1.5.5 Servicing line insulation. The liquid servicing line, valve, filter, and fittings outside the tank assembly outer shell shall be insulated with type II, class 1, pipe covering conforming to HH-I-551. The insulation shall be so applied, covered, and sealed that entrance of moisture will be prevented.

3.8.1.8.1.6 Liquid servicing hose. One 10-foot length of cryogenic liquid hose, part No. 59C6671-2 conforming to MIL-H-27301, shall be provided with the tank. One end of the hose shall be connected to the 1-inch male pipe thread of the liquid servicing line. The other end of the hose shall be provided with a 1-inch close nipple and a 1-inch, female, liquid-oxygen coupling half assembly consisting of a coupling cone, coupling nut, retaining ring, and plug conforming to Drawings C8987856, C8987857, B8987839, and C8987870, respectively, assembled as shown on Drawing C8987831.

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3.8.1.8.1.7 Vapor vent line. The vapor vent line shall permit filling the liquid storage tank assembly through the liquid fill-drain line specified in 3.8.1.8.1.1.1 at any rate up to and including 100 gpm without the inner shell vapor phase pressure exceeding 15 psig when the liquid storage tank assembly is cold. The vapor vent line shall also permit cooling down of the liquid storage tank assembly within 15 minutes when all components have stabilized at 125°F without the inner shell vapor pressure exceeding 15 psi. The vapor vent line shall be of not less than 2 inches diameter. The vapor vent line installation shall be such as to minimize heat transfer between the inner and outer shells and to provide for filling the tank to not less than 2,040 gallons before liquid vents through the vapor vent line.

3.8.1.8.1.7.1 Vapor vent line shutoff valve. A manually operated shutoff valve having a free passage size approximately equal to that of the vapor vent line inner diameter when fully open shall be provided in the vapor vent line as near as practicable to the point where it enters the outer shell. The valve shall provide positive shutoff when closed. General design, construction, and mounting shall be as specified in 3.8.1.10.1 through 3.8.1.10.2.

3.8.1.8.1.7.1.1 Vapor vent line shutoff valve mounting. The vapor vent line shutoff valve shall be solidly mounted in a location readily accessible to an operator standing on the ground in front of the control panel. The vapor vent line shutoff valve shall be so mounted to supporting members that force applied in opening or closing the valves will not damage the vapor vent line piping.

3.8.1.8.1.7.2 Vapor vent line discharge. The vapor vent line shall discharge toward the ground outside the control housing assembly in a location where high concentrations of oxygen gas and liquid overflow will not constitute a hazard or create fog in the control housing assembly. The line shall terminate in a 2-inch female pipe thread in an area which is convenient for attachment of a hose from a purging unit.

3.8.1.8.1.8 Liquid level gage lines. If a differential-pressure-type liquid level gage is furnished, separate liquid level gage sensing lines to the inner shell shall be provided and shall be of the minimum size consistent with dependable operation of the gage.

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3.8.1.9 Safety devices. Where necessary, pressure relief devices shall be provided for protection of the liquid storage tank assembly and its associated components. All pressure relief devices and drains shall be installed in readily accessible locations and, except for the outer shell rupture disc, shall discharge all vented liquid or vapor toward the ground in a location which will insure against (a) damage to the equipment or danger to personnel, and; (b) creating fog where it will obscure controls, instruments, or fittings in the control housing assembly. All safety devices shall be fabricated from corrosion-resistant materials not adversely affected by extended contact with oxygen, moist air, water, or cleaning solvents such as trichloroethylene.

3.8.1.9.1 Inner shell safety head. A bursting-disc-type safety head utilizing a readily replaceable disc in a union-type fixture shall be provided to relieve the inner shell vapor phase pressure to the atmosphere should this pressure become excessive and the pressure relief valve fail to open. The safety head relief pressure shall be not less than 1-1/2 times the tank inner shell maximum working pressure.

3.8.1.9.1.1 Inner shell safety head installation. The inner shell safety head shall be installed with a sufficient length of tubing between itself and the vapor vent line to prevent frosting of the safety head unless the safety head actually ruptures to relieve the inner shell pressure.

3.8.1.9.2 Outer shell safety head. A bursting-disc-type safety head or a flat-plate-type safety disc assembly in which the plate bears on an O-ring shall be provided on the outer shell for venting the annular insulation space to the atmosphere should the annular insulation space pressure become excessive. If a bursting-disc-type safety head is provided, it shall be equipped with a vacuum support to prevent inward distortion when the annular insulation space is evacuated. The flat-plate-type safety disc assembly may be used only if the plate will be retained against the O-ring with sufficient pressure to provide a positive seal against the entrance of gases or moisture under all conditions of operation, even when no vacuum exists in the annular insulation space.

3.8.1.9.2.1 Outer shell safety head installation. The outer shell safety head shall be installed on the end of the tank farthest removed from the control housing assembly. Design and installation shall be such that discharge from the safety head will be directed away from the control housing assembly and personnel adjacent thereto.

3.8.1.9.3 Inner shell vapor space pressure control. A pressure control device shall be connected into the liquid storage tank assembly vapor vent line between the outer shell and the vent line shutoff valve to vent gas as necessary to prevent the inner shell vapor space pressure exceeding the set point by more than 2-1/2 psig. The device shall not vent gas until the inner shell vapor pressure rises to within 2-1/2 psig of the set pressure. The pressure venting set point shall be adjustable by means of a control knob from 10 through 50 psig.

3.8.1.9.3.1 Pressure control device installation. The pressure control device shall be installed in a manner that will prevent freezeup of the device when

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venting oxygen vapor and will permit the inner shell vapor space to be vented to atmospheric pressure by opening the vent valve.

3.8.1.9.3.1.1 Inner shell vapor space pressure relief valve. An automatic pressure relief valve shall be connected into the vapor vent line between the outer shell and the vent line shutoff valve to relieve inner shell excess pressure. The relief valve shall open at a pressure sufficiently below the inner shell bursting-disc-type safety head breaking pressure to insure against rupture of the safety head unless the relief valve fails. The opening pressure shall be not less than 55 psig, and the valve shall be sized to insure against the inner shell vapor space pressure exceeding 65 psig when:

(a) The tank is filled to not less than 90 percent of design capacity

(b) The buildup valve is left wide open

(c) All other valves are closed.

3.8.1.9.3.1.1.1 Inner shell vapor space pressure relief device installation. The inner shell vapor space pressure relief valve shall be installed in a manner that will prevent frosting of the relief valve unless the relief valve actually opens to relieve the inner shell vapor phase pressure.

3.8.1.9.4 Liquid fill-drain line pressure relief and drain valves

3.8.1.9.4.1 Fill and drain line pressure relief valve. An automatic pressure relief valve shall be provided in the fill-drain line between the shutoff valve and the disconnect coupling for venting excess pressure from the fill-drain line. The relief valve shall be designed to open at a pressure in excess of that at which the inner shell vapor space pressure relief valve opens but less than the safe working pressure of the fill-drain line.

3.8.1.9.4.1.1 Fill-drain line automatic pressure relief valve installation. The fill-drain line automatic pressure relief valve shall be installed with a sufficient length of tubing between itself and the fill-drain line to prevent frosting of the relief valve unless the relief valve actually opens to relieve the fill-drain line pressure.

3.8.1.9.4.2 Fill-drain line drain valve. A manually operated valve for draining liquid from the fill-drain line disconnect coupling when removing the fill-drain hose shall be provided in the fill-drain line between the shutoff valve and the disconnect coupling, as near the disconnect coupling as practicable. The drain valve shall be located and installed to be readily accessible to an operator standing on the ground outside the control housing assembly. If practicable, the automatic pressure relief valve and the drain valve may be combined into one unit. The drain valve shall discharge beneath the control housing assembly.

3.8.1.9.5 Liquid servicing line pressure relief and drain valves

3.8.1.9.5.1 Servicing line pressure relief valve. An automatic pressure relief valve shall be provided in the servicing line between the shutoff valve and the servicing hose for venting excess pressure from the servicing line and servicing hose. The relief valve shall be designed to open at a pressure in excess of that at which the inner shell vapor space pressure relief valve opens but less than the safe working pressure of the servicing line and servicing hose.

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3.8.1.9.5.1.1 Servicing line automatic pressure relief valve installation. The servicing line automatic pressure relief valve shall be installed with a sufficient length of tubing between itself and the servicing line to prevent frosting of the relief valve unless the relief valve actually opens to relieve the servicing line pressure.

3.8.1.9.5.2 Servicing line drain valve. A manually operated valve for draining liquid from the servicing line and the servicing hose shall be provided in the servicing line between the shutoff valve and the servicing hose as near the servicing hose as practicable. The drain valve shall be located and installed to be readily accessible to an operator standing on the ground outside the control housing assembly. If practicable, the automatic pressure relief valve and the drain valve may be combined into one unit. The drain valve shall discharge beneath the control housing assembly.

3.8.1.9.6 Additional pressure relief valves. Additional automatic pressure relief valves shall be installed in each circuit or component not specified and which can be so isolated by closing valves or otherwise that a dangerous pressure could build up. These relief valves shall each be designed to open at a pressure sufficiently low to insure against damage to the component or circuit affected. However, the operating pressure shall be in excess of that at which the inner shell vapor space pressure relief valve opens.

3.8.1.10 Controls

3.8.1.10.1 Shutoff and control valves. Manually operated shutoff and control valves shall be provided wherever needed for control of the liquid storage tank assembly inner shell vapor phase pressure, vapor phase venting, liquid transfer, and tank instrumentation.

*3.8.1.10.1.1 Valve design and construction. Valves shall be designed and constructed to provide positive shutoff when closed and minimum resistance to flow when open. The globe valves of 1/2 inch and larger nominal size handling low temperature liquid or gas shall incorporate renewable seats that can be removed and replaced without disassembling the valve body from the tank piping. The sealing discs or plugs for these valves shall also be renewable without replacing the valve stem. All valves handling low temperature liquid or gas shall be so designed or modified that the packing gland is located not less than the distance specified in table II from the valve connection centerline. The stem and the stem housing on these valves shall either be fabricated from 18-8 stainless-steel or other equally low heat conductive material, or the stem and stem housing shall be provided with 18-8 stainless-steel or other equally low heat conductive material inserts equal in length to not less than one-half the distance specified in table II. Liquid level gage control and other instrument control and isolation valves may be of conventional construction.

3.8.1.10.1.1.1 Valve handles. Each manually operated control or shutoff valve shall be provided with a tee, cross, or other easily turned type handle with a diameter equal to 3 inches per inch of nominal size, but not less than 1 inch.

3.8.1.10.1.1.2 Valve cycling and leakage performance. When closed with a torque of 60 ±5 pound-inches per inch of nominal size, and with the valve body at either ambient temperature or the atmospheric

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TABLE II. Separation Of Valve Packing Gland From Valve Connection Centerline

Valve Nominal Size (in Inches)	Separation Of Packing Gland From Connection Centerline in Inches
1/4	4
3/8	4
1/2	7
3/4	7
1	9
1-1/4	9
1-1/2	11
2	11
2-1/2	12
3	12

pressure boiling temperature of liquid oxygen, the valves shall not leak more than 2 cubic inches of free air, oxygen, or nitrogen gas per hour per inch of nominal size from 50 psig to a downstream pressure of 0 psig. The valves shall be capable of not less than 2,000 cycles of operation (see 6.3.6) when subjected to a differential pressure of 50 psig while closed, including at least 1,000 cycles at ambient temperature and 1,000 cycles with the valve body at the atmospheric pressure boiling temperature of liquid oxygen, without adjustment, repair, or maintenance, and without the above specified leakage rate being exceeded before, during, or after the cycling.

3.8.1.10.1.1.3 Valve overtorque performance. The valves shall be designed and constructed to withstand a closing torque of 300 ±10 pound-inches per inch of nominal size with the valve bodies at either ambient temperature or the atmospheric pressure boiling temperature of liquid oxygen. Such operation shall not result in a parts failure, or a leakage rate in excess of that specified in 3.8.1.10.1.1.2 when subsequently closed with a torque of 60 ±5 pound-inches per inch of valve nominal size.

3.8.1.10.1.2 Valve material. The valves shall be fabricated from corrosion-resistant material not adversely affected by extended contact with liquid or gaseous oxygen or nitrogen, moist air, water or cleaning solvents such as trichloroethylene.

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3.8.1.10.1.3 Valve packing. Valve packing shall be of a material safe for use in contact with liquid or gaseous oxygen or nitrogen and that will effectively seal the valve stem without causing difficulty of operation.

3.8.1.10.2 Valve mounting. All valves used during liquid transfer shall be so mounted and installed that they can be readily opened or closed, and so that their control handles will be readily available to an operator standing on the ground at the control panel. All valves shall be so mounted to supporting members or large size piping that force applied to the control handles in opening or closing the valves will not damage the tank piping, and so mounted that the operator will be protected from liquid spray if a fitting or coupling leaks or breaks.

3.8.1.11 Instruments. All instruments necessary for effective and safe operation of the tank shall be provided. These shall include the following:

3.8.1.11.1 Liquid level gage. A liquid level gage designed for flush mounting on the control panel and having a dial diameter of not less than 6 inches shall be provided to indicate the quantity of liquid oxygen contained in the liquid storage tank assembly inner shell with an accuracy of ±3 percent of tank design capacity.

3.8.1.11.1.1 Gage construction. If a differential-pressure-type liquid level gage is furnished, the gage design and construction shall be such that breakage of the glass face will not subject the sensing element to damaging pressures when the inner shell is pressurized to 55 psig.

3.8.1.11.1.2 Graduations. The liquid level gage shall be graduated in gallons of liquid oxygen. The liquid-oxygen graduations shall be printed on the gage face in black on a white background.

3.8.1.11.1.3 Liquid level gage mounting. The liquid level gage shall be securely mounted on the control panel in a position where it will be readily visible during liquid transfer or filling to an operator standing on the ground in front of the control panel.

3.8.1.11.1.4 Pulsation snubbers. If a differential-pressure-type liquid level gage is furnished, it shall be provided with restriction orifices in each sensing line to act as pulsation snubbers for preventing undue fluctuation of the indicating needle.

3.8.1.11.2 Vapor phase pressure gage. A pressure gage with a dial diameter of not less than 6 inches shall be provided for indicating the inner shell vapor phase pressure. The gage shall be designed for flush mounting, shall be graduated over approximately 270 degrees of the dial face, and shall be accurate to within ±2 percent of full-scale range.

3.8.1.11.2.1 Range. The indicating range of the vapor phase pressure gage shall be approximately 1-1/2 times the normal working pressure of the inner shell, unless a higher range is necessary to prevent the indicator from moving off scale before reaching the relief pressure of the inner shell safety head and the vapor space pressure relief valve specified in 3.8.1.9.1 and 3.8.1.9.3.

3.8.1.11.2.2 Vapor phase pressure gage installation. The vapor phase pressure gage shall be so installed that no combination of valve or control settings can

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interfere with its indicating the inner shell vapor space pressure. The gage shall be securely mounted on the control panel in a position where it will be readily visible during liquid storage or transfer to personnel standing on the ground in front of the control panel.

3.8.1.11.2.3 Pressure sensing tap. A tap into the pressure gage line shall be provided for attachment and use of external test equipment. The tap shall terminate in a -4 size flared tube connection conforming to MS33656 installed on the control panel near the pressure gage. The fitting shall be sealed with an AN929-4 cap assembly.

3.8.1.11.3 Vacuum indicator. A vacuum indicator shall be provided to indicate the actual absolute pressure within the liquid storage tank assembly annular insulation space.

3.8.1.11.3.1 Type. If electrically operated, the vacuum indicator shall be of the intermittent-reading type having externally attached batteries. It shall be wired to consume current and indicate the absolute pressure only when the reading switch is closed. The reading switch shall be spring-loaded or otherwise constructed to require physical pressure for closing and to automatically open when the pressure is released.

*3.8.1.11.3.2 Vacuum connection. Connection to the annular insulation space shall be made through a filter that will prevent insulating material from entering the gage sensing element. Provisions shall also be incorporated that will permit replacing the sensing element without destroying or appreciably degrading the vacuum in the annular insulation space.

3.8.1.11.3.3 Mounting. The vacuum indicator and the reading switch shall be mounted on the control panel where they will be readily visible and accessible to an operator standing on the ground in front of the control panel.

3.8.2 Pressure buildup system. The pressure buildup system shall operate to pressurize the liquid storage tank assembly inner shell vapor space to 30 psig when the tank contains 100 gallons or more of liquid oxygen.

3.8.2.1 Buildup rate. The pressure buildup system shall pressurize the inner shell vapor space to 30 psig within 10 minutes after closing the vapor vent line shutoff valve and initiating buildup when the inner shell is filled to between 15 and 20 percent of rated capacity and has been vented to atmospheric pressure for not less than 4 hours.

3.8.2.2 Pressure buildup system components. The pressure buildup system shall consist of a vaporization coil and controls, together with the connections, piping, mountings, instruments, and accessories needed for proper and safe operation.

*3.8.2.3 Method of operation. The pressure buildup system shall operate by vaporizing a portion of the liquid contained in the liquid storage tank assembly inner shell and pressurizing the inner shell vapor space with the resultant vapors. Vaporization of the liquid shall be accomplished in an atmospheric air heated coil that forms a direct external connection between the inner shell liquid and vapor phases through the pressure control system.

*3.8.2.4 Vaporization coil. The vaporization coil shall be designed and constructed to receive liquid from the liquid storage tank assembly inner shell and

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vaporize it at the rate necessary for compliance with the performance requirements of 3.8.2.1 by heat exchange with ambient air, and discharge the resultant vapors into the inner shell vapor space. The vaporization coil shall include all shrouding, baffling, and ducting necessary for insuring proper airflow distribution over the heat exchange surfaces.

3.8.2.4.1 Vaporization coil material. The vaporization coil and the associated piping, supports, attachments, and connections shall be constructed from materials of types that will not be attacked or adversely affected by contact with liquid oxygen, gaseous oxygen, water, dust, mud, moist air, or cleaning solvents.

3.8.2.4.2 Vaporization coil installation. The vaporization coil shall be installed where it will be protected from adverse environmental conditions and mechanical damage. The connection of the vaporization coil to the vent line shall be located in a readily accessible area and designed to permit easy disconnection of the vaporization coil from the vent line. A cap shall be provided to seal the hole in the vent line resulting from disconnection of the vaporization coil.

*3.8.2.5 Operating controls. The pressure buildup system shall include all controls necessary for complying with 3.8.2 and 3.8.2.1. Normal operation of the pressure buildup system shall require no control operation other than closing of the vapor vent line valve and opening of the pressure buildup control valve.

3.8.2.5.1 All manually actuated controls used in normal operation of the pressure buildup system shall either be mounted on the control panel or easily accessible within the control housing assembly.

3.8.3 Control housing assembly. A sturdy, heavy-gage, sheet-metal housing assembly shall be fabricated onto one end of the liquid storage tank assembly. The housing shall contain and provide protection for the tank controls, for the tank instrumentation, and for the accessories.

3.8.3.1 Control housing assembly design and construction. The control housing assembly shall be designed and constructed to protect the enclosed components from mechanical damage, direct solar radiation, rain, sleet, snow, mud, sand, and other adverse weather and environmental conditions. The control housing assembly shall be so constructed that it, or applicable sections thereof, can be readily removed from the liquid storage tank assembly for maintenance or removal of components.

3.8.3.1.1 The control housing assembly shall be constructed from material resistant to corrosion caused by wetting with condensed water, and so that retention of condensed water within the control housing will be effectively prevented.

3.8.3.1.2 If practicable, the top of the control housing assembly and the tops of all protuberances therefrom shall be shaped and slanted to resist the retention of snow, rain, or sleet. Doors and panels that must be opened for use of the tank shall be so constructed and installed that snow or ice will not lodge thereon in a manner that will prevent easy operation or access.

3.8.3.2 Size. The control housing assembly shall be of sufficient size to provide ready access to the enclosed components for operation, servicing, maintenance, and repairs, and to provide space for the storage of accessories, liquid servicing hose, and tools.

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3.8.3.3 Access openings. Doors or hinged panels shall be provided in the control housing assembly as necessary for access for operation, inspection, servicing, maintenance, and repair of the equipment. The doors and hinged panels whose positions may have to be regulated shall be provided with means for securing them in any position from fully closed to wide open when subjected to wind velocities up to and including 70 miles per hour (mph).

3.8.3.3.1 Hinges and latches. All doors and hinged panels shall be equipped with sturdy hinges and with latches that will hold them securely closed during storage. Latches shall also be provided for securely holding open the doors and panels that must be opened for tank use. These latches shall securely retain the doors or panels in the set position even when subjected to wind gusts up to and including 70 mph.

3.8.3.3.2 Latches. The latches and position regulating devices on all doors and hinged panels that must be opened for normal operation, servicing, and maintenance of the tank shall be of the quick-release type, readily operable by personnel wearing heavy arctic mittens and without tools. Provisions shall be incorporated for locking the control housing assembly with common padlocks.

3.8.3.4 Ventilation provisions. Ventilation openings or other air circulation provisions shall be provided in the control housing assembly as necessary to prevent the accumulation of gases, or of fog resulting from moist air being chilled in the control housing during liquid transfer or tank filling. The ventilation openings shall be so located or shielded that the entrance of rain, snow, or other precipitation will be effectively prevented.

3.8.3.5 Liquid servicing hose storage provisions. Provisions for protection and storage of the liquid servicing hose shall be furnished inside the control housing assembly. The storage provisions shall include quick-disconnect fasteners which will securely retain the servicing hose in proper storage position while it is connected to the servicing line and the control housing assembly access openings are closed. The storage provisions shall protect the hose against damage when the tank is subjected to the acceleration forces specified in MIL-A-8421. The free end of the servicing hose shall be covered to protect it against entrance of contaminants. The hose shall be easily and quickly released for servicing by personnel wearing heavy gloves.

3.8.4 Control panel. A control panel shall be provided and shall be of sufficient size for installation of the instruments and controls needed in normal tank use. It shall be so installed on the tank that all instruments, controls, et cetera, mounted thereon will be readily visible and accessible to an operator standing on the ground in front of the control panel. All valve handles and gage faces shall be not less than 27 inches above ground level. Whenever practicable, instruments that must be monitored in making a control adjustment shall be so located that an operator or maintenance man can simultaneously make the control adjustment and adequately read the related instrument.

3.8.4.1 Control panel protection. The control panel shall be protected by a panel or door when the tank is not in use for liquid transfer. The control panel shall be so mounted or recessed into the control housing assembly that it will be shielded from rain and snow when the door or panel is open.

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3.8.4.2 Instruments and controls

3.8.4.2.1 Instruments. All instruments necessary for indicating the information needed in controlling the equipment and determining its performance shall be securely mounted on the control panel.

3.8.4.2.2 Controls. Insofar as practicable, all manually operated controls used in normal operation of the equipment shall be located on the control panel.

3.8.5 Skid base. The liquid storage tank assembly shall be provided with a skid base designed to serve as a combination base and mounting platform for the tank when it is placed on a flat, level, hard surface such as a concrete floor or pad. Overall size of the skid base shall be such as to insure distributing the storage tank weight over an area not less than the vertical projection of the outer shell outline.

3.8.5.1 Bearing members. The skid base members bearing on the supporting surface shall have a width of not less than 5 inches in contact with the supporting surface. The bearing members shall include not less than one member on each side of the tank, each member constructed to be in contact with the supporting surface over not less than the length of the outer shell and the control housing assembly.

*** 3.8.5.2 Skid base ends. Each end of the skid base shall be curved with a radius of not less than 6 inches or turned up at a 30- to 45-degree angle so that the bottom surface rises smoothly to a height of not less than 6 inches above a flat surface on which the tank is placed.

*** 3.8.5.3 Towing provisions. The skid base shall be designed and fabricated to permit and facilitate attachment of a cable and pulling the tank over a flat concrete or asphalt paved surface by a tug or other towing vehicle without distortion of or damage to the tank.

*** 3.8.5.4 Forklift entries. The skid base shall incorporate symmetrically about the cg centerline of the empty tank the forklift entries shown on figure 3. The entries shall extend the complete width of the skid base and shall contain no obstructions or irregularities that could impede movement of forklift tines through the length of the entries. The entries shall restrain 72-inch or shorter forklift tines from tilting or oscillation as necessary to prevent damaging the tank during lifting or movement operations.

3.8.6 Special operation and maintenance tools. When specified (see 6.2), a complete set of all special tools other than common handtools required for proper operation and maintenance of the tank and its accessories shall be provided.

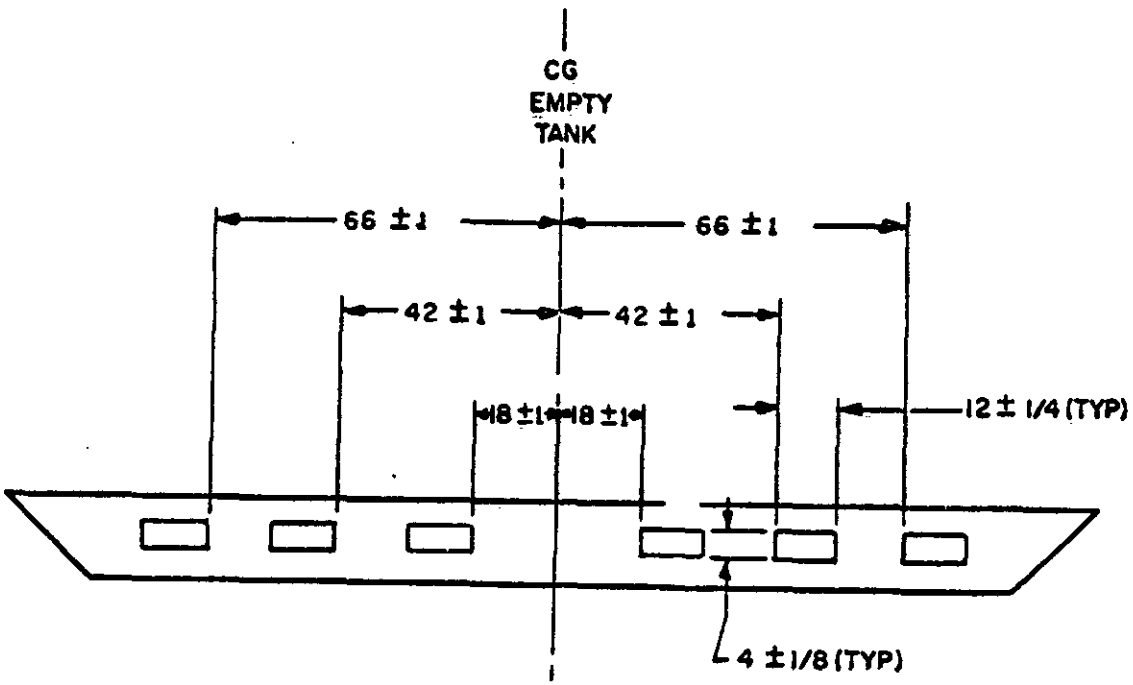
3.8.7 Compartments

3.8.7.1 Tool compartment. A compartment for storage of all tools required in normal operation and maintenance of the tank shall be provided. Provisions shall be incorporated for locking the tool compartment with a common padlock.

3.8.7.1.1 Couplings. Two 3-1/2-inch, female liquid-oxygen coupling half assemblies consisting of a coupling cone, coupling nut, retaining ring, and plug conforming to Drawing C8984435, C8984434, B8987839, and C8984991, respectively, assembled as shown on Drawing C8987831, shall be supplied in the tool compartment for use in adapting the fill-drain line to nonstandard supply sources.

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2,000 GALLON CAPACITY
LIQUID OXYGEN
STORAGE TANK



NOTE:

DIMENSIONS IN INCHES

TOLERANCES AS NOTED

FIGURE 3. Forklift and Entries

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3.8.7.2 Handbook and record compartment. A compartment shall be provided integral to each end item for containing the equipment operation handbook and the equipment maintenance and inspection record. The compartment shall be weatherproofed if not contained in an enclosure and a minimum size of 9 by 11-1/2 by 2 inches.

3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern manufacturer's part numbers and changes thereto.

3.10 Dimensions. The overall dimensions of the tank shall not exceed the following when all access openings are closed:

	Feet
Length	16
Width	8
Height	9

3.11 Weight. The weight of the empty tank and its accessories shall not exceed 12,000 pounds.

3.12 Finishes and protective coatings

3.12.1 Surfaces contacting oxygen. All surfaces, parts, fittings, et cetera, of the tank that will be in contact with high-purity oxygen shall be thoroughly cleaned with a nonflammable solvent, such as trichloroethylene, to remove all metal shavings, oil, grease, and other foreign material. No other cleaning, priming, or painting with organic materials shall be performed on these surfaces.

3.12.2 Exposed parts and surfaces. All exposed metal parts and surfaces except parts and surfaces that contact high-purity oxygen, shall be cleaned, treated, and finished as specified in MIL-STD-808. When more than one finish process is permitted by MIL-STD-808 for a part or surface, the most applicable finish process shall be determined by the contractor unless otherwise specified herein.

3.12.2.1 Finish code FF-909 shall apply to all exterior parts and surfaces of the tank.

★

3.12.2.2 Final film color designation YL shall apply to control panels, equipment items, and mounting parts installed within the control housing assembly.

3.13 Operational markings

3.13.1 Tank marking. Each tank shall be marked as follows by applying reverse silk-screened reflectorized tape conforming to L-S-300:

3.13.1.1 The lettering LIQUID OXYGEN shall be applied in 6-inch green block-type letters on each side of the tank and at both ends in approximately the center.

3.13.1.2 The lettering NO SMOKING WITHIN 50 FEET shall be applied in a single line with 4-inch red reflective letters. The marking shall be applied on each side of the tank and at both ends below the lettering LIQUID OXYGEN.

3.13.2 Control marking. All valves, gages, controls, and indicators used in operation of the tank shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not fade or cause them to become eradicated. Tags shall not be used.

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* 3.13.3 Warning plate. A weather proofed red plate containing the following information shall be provided adjacent to the liquid servicing line:

WARNING — TANK IS NOT TO BE FILLED THROUGH THIS LINE — FOR SERVICING ONLY

3.13.4 Operating and precautionary instructions. Brief operating and precautionary instructions shall be permanently affixed on or near the tank control panel. The instructions shall be clear, concise, and adequate to enable operation of the tank without damage to the equipment or injury to personnel. Instruction panels shall be made from sheet aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised markings with a black or other dark color background, and with a border of not less than 1/4 inch.

3.13.5 Transportation data plate. A transportation data plate in accordance with MIL-P-514 shall be securely attached to the outside of the control housing assembly adjacent to the nameplate. The transportation data plate shall include the following information:

TRANSPORTATION DATA PLATE

Shipping Weight	Overall Width
Shipping Cubage	Overall Height
Overall Length	C. G. Location

T. O. or T. M. Number

* 3.13.6 Lifting instruction plate. An instruction plate shall be furnished containing all information necessary for transportation personnel to:

(a) Safely lift and move the empty tank with a single, high capacity, forklift truck

(b) Safely lift and move the empty tank with multiple, smaller (down to 6,000 pounds) capacity, forklift trucks, and

(c) Rig a safe lifting sling from common cable and spreader bar components and safely lift the empty tank with a 2-g acceleration.

The plate shall be securely attached to the outside of the control housing assembly near the transportation data plate. The lifting instruction plate shall be of the same material and prepared in the same manner as the transportation data plate.

3.13.7 Color marking. The indicating scale of each instrument (pressure gage, differential pressure indicator, temperature indicator, liquid level indicator, et cetera) used for normal operation and control of the tank shall be permanently and plainly marked with green and red to show the proper operating and the other-than-proper-operating or danger zones respectively.

3.13.8 Insulation identification plate. An insulation identification plate containing all information necessary to completely identify the insulation material and showing the quantity in pounds of the insulation material installed in the liquid storage tank assembly annular insulation space shall be permanently affixed in a readily accessible and protected location that will be within the control housing assembly when it is closed for transport or storage. The insulation identification plate shall be prepared in the same manner as the operation and precautionary instruction plate.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.14.1 Tank nameplate location. The tank nameplate shall be securely attached to the outside of the control housing assembly in a readily visible location.

3.15 Workmanship. All parts of the tank shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to the following:

(a) Freedom from blemishes, defects, burrs, and sharp edges

(b) Accuracy of dimensions, radii of fillets, and marking of parts and assemblies

(c) Thoroughness of soldering, welding, brazing, painting, and riveting

(d) Thorough removal of rust, slag, scale, flux, and other foreign materials from the inside of the liquid storage tank assembly and from all piping, valve, connection, filter housing and other surfaces that contact oxygen product

(e) Alignment of parts and tightness of assembly screws, bolts, rivets, et cetera

(f) That rivets are tight and properly headed.

3.16 General cleaning instructions. Following completion of fabrication and assembly operations, the tank shall be thoroughly cleaned and degreased to remove dirt; excess soldering, brazing, and welding flux; welding slag; scale; loose, spattered, or excess solder; metal chips; loose or chipped paint; spilled chemicals; and other foreign materials. Cleaning shall be continued until visual inspection shows no evidence of contamination or foreign matter, ultraviolet inspection of surfaces contacting oxygen product shows no evidence of fluorescence, and particulate contamination of liquid oxygen withdrawn does not exceed the following:

(a) When the tank is filled to 90 percent or more of design capacity with filtered liquid oxygen and allowed to set for not less than 2 hours, the residue remaining after evaporation of the first liter of liquid withdrawn through the liquid fill-drain line with its filter removed shall contain:

(1) No solid particle with any dimension greater than 1,000 microns

(2) No fibrous particle (see 6.3.7) with a length greater than 6,000 microns

(3) No more than 25 milligrams total of both solid and fibrous particles

(b) When the liquid oxygen remaining in the tank after withdrawal of the 1-liter sample specified above is discharged through the fill-drain line specified in 3.8.1.8.1.1 and the filter specified in 3.8.1.8.1.4 at not less than 90 gpm, the total amount of contaminant collected on the filter shall not exceed 0.25 gram.

3.16.1 Degreasing. The tank surfaces parts, fittings, et cetera, that will be in contact with high-purity oxygen shall be degreased by one of the following methods. Precautions shall be taken to insure that solvents do not contact parts fabricated from incompatible materials.

Method A — Vapor degreasing process, using type II stabilized trichloroethylene conforming to O-T-634 in a standard commercial vapor degreaser or by blowing degreasing vapors into the component in such a manner that the vapor will condense on and properly clean all surfaces requiring degreasing. Operation of a commercial vapor degreaser shall be

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in accordance with the manufacturer's recommendations. Following the vapor degreasing treatment, all solvent shall be removed by baking in an oven; by purging with hot, dry oil-free air or nitrogen; or by vacuum evacuation.

Method B — Solvent degreasing process, using stabilized trichloroethylene or commercial oxygen-safe cleaning solvent at ambient temperatures to thoroughly wash all surfaces requiring degreasing. If the solvent contains more than 1 percent oil or hydrocarbons after the cleaning and being drained from the component, the degreasing operation shall be repeated with clean solvent. Following the degreasing process, all solvent shall be removed by baking in an oven; by purging with hot, dry, oil-free, air or nitrogen; or by vacuum evacuation.

Method C — Detergent degreasing process, in which the components to be cleaned are washed with hot inhibited alkaline cleaner until free from oil, grease, and other contaminant materials. Following this treatment, all surfaces (internal and external) shall be rinsed thoroughly with fresh, clean, hot water and dried by blowing with filtered and dried oil-free air or nitrogen, or by baking at a temperature of 250° to 300°F until all water is removed.

3.16.1.1 Petroleum and other flammable solvents shall not be used on such surfaces.

3.16.1.2 All cleaning materials and solvents shall be thoroughly removed from the tank components following the cleaning and degreasing processes.

* 3.17 Lubrication. All tank machined bearing surfaces that are normally lubricated in operation and that do not come into contact with the contained product shall be thoroughly lubricated with the recommended military specification or federal specification lubricants to be ready for immediate operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the tank shall be classified as follows:

(a) Preproduction testing (4.4)

(b) Acceptance tests (4.5)

4.3 Test conditions

4.3.1 Tools. Special tools and field equipment shall be used to the maximum practicable extent during testing. Use of these tools shall be sufficient to determine their usefulness. Instances where the special tools furnished are inadequate shall be recorded in detail.

4.3.2 Instrumentation

4.3.2.1 Pressures. Pressures in excess of 1 millimeter (mm) Hg absolute and pressure differentials, except pressure drops in the fill-drain and servicing

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line, shall be measured by means of the pressure gages and differential pressure meters furnished as part of the tank.

4.3.2.1.1 Absolute pressures below 1 mm Hg shall be measured by the use of calibrated instruments designed to read total pressures (including permanent gages and condensible vapors).

4.3.2.1.2 Pressures in excess of atmospheric shall be reported in pounds per square inch gage. Pressures up to and including atmospheric shall be reported in microns or millimeters of mercury absolute, as applicable. Differential pressures shall be reported in inches of water or inches of mercury, as applicable.

4.3.2.1.3 Barometric pressure. Barometric pressure shall be measured by a properly calibrated mercurial barometer and reported in millimeters of mercury absolute.

4.3.2.2 Temperatures. Temperatures shall be measured by appropriately located thermometers, or by thermocouples used with calibrated potentiometers. Temperatures shall be recorded in degrees Fahrenheit.

4.3.3 Flow rates

4.3.3.1 Liquid. Liquid oxygen flow rates shall be determined by either of the following methods:

(a) Passing the liquid through a properly calibrated recording or totaling-type liquid flowmeter designed to handle the liquid being measured

(b) Collecting the liquid and weighing on an accurate scale

4.3.3.1.1 Liquid flow rates shall be reported in gallons per minute of the liquid at its atmospheric pressure boiling temperature.

4.3.3.2 Gas. Gaseous oxygen flow rates shall be determined by passing the material through a properly calibrated recording or totaling-type gas meter designed to handle the gas being measured.

4.3.3.2.1 Gaseous flow rates shall be converted to weight flow rates for the material handled, expressed in pounds per hour, pounds per minute, et cetera, as applicable.

4.3.4 Observations. During the progress of all tests, such characteristics as ease of control, pressure regulation, freedom from leaks, and ease of operation shall be observed to determine whether the tank complies with all requirements specified herein.

4.3.5 Vacuum pumping. A vacuum pump shall be attached to the tank vacuum line disconnect fitting and the liquid storage tank annular space pumped down to insure an absolute pressure of less than 100 microns Hg within 24 hours after the warm liquid storage tank assembly is filled with liquid oxygen to 40 percent or more of design capacity.

4.3.5.1 Vacuum line seal. Upon completion of the vacuum pumping, the vacuum pump shall be disconnected and the vacuum line so sealed that further vacuum pumping cannot be accomplished without breaking the seal. The vacuum line shall remain sealed throughout all testing specified herein.

4.3.6 Test data

4.3.6.1 Correction of data. If the tests specified herein cannot be conducted under

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the specified conditions, the tests may, upon approval of the procuring activity, be conducted under other conditions and the performance under the specified conditions calculated from the test results obtained. The manner of calculation shall be demonstrated, and actual test data proving the correctness of the calculation methods shall be presented for review by the procuring activity.

4.3.6.2 Operational data. Operational data shall be recorded at the intervals specified under the individual test instructions. The data recorded shall include the following:

(a) Liquid storage tank

(1) Tank vapor phase pressure (psig)

(2) Tank liquid level gage reading (gallons)

(b) Servicing data

(1) Pressure of liquid entering transfer hose (psig)

(2) Flowmeter reading (gpm)

(3) Pressure of liquid at end of transfer hose (psig)

(4) Actual delivery rate of liquid (gpm).

4.3.6.3 Psychrometric data. Wet- and dry-bulb temperature readings shall be recorded at the intervals specified under the individual tests.

4.3.6.4 Barometric pressure. The barometric pressure shall be measured and recorded at the intervals specified under the various test procedures specified under 4.6.

4.3.6.5 Tolerances

4.3.6.5.1 Data on absolute pressures measured in the range from 0 to 10 microns Hg shall be accurate to within 0.5 micron Hg.

4.3.6.5.2 Data on absolute pressures measured in the range from 10 to 50 microns Hg shall be accurate to within 2 microns Hg.

4.3.6.5.3 Data on absolute pressures measured in the range from 50 to 250 microns Hg shall be accurate to within 5 microns Hg.

4.3.6.5.4 Data on absolute pressures measured in the range from 250 to 500 microns Hg shall be accurate to within 10 microns Hg.

4.3.6.5.5 Data on absolute pressures measured in the range from 500 to 1,000 microns Hg (1mm Hg) shall be accurate to within 20 microns Hg.

4.3.6.5.6 Data on absolute pressures measured in the range from 1 mm Hg to and including atmospheric shall be accurate to within 0.1 mm Hg.

4.3.6.5.7 Data on gage pressures measured in the range from atmospheric to 10 psig shall be accurate to within 1/2 psig.

4.3.6.5.8 Data on gage pressure measured in the range from 10 to 50 psig shall be accurate to within 1 psig.

4.3.6.5.9 Data on gage pressures measured in excess of 50 psig shall be accurate to within 2 percent of the numerical reading.

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4.3.6.5.10 Data on weights, weight flow rates, and volume flow rates shall be accurate to within 1 percent of the numerical reading.

4.3.6.5.11 Data on gas and vapor flow rates shall be accurate to within 1 percent of the numerical reading.

4.3.6.5.12 Data on temperatures shall be accurate to within 2°F.

4.3.6.6 Pressure test record. A record of the pressures at which each tank component is tested and the length of time during which it is subjected to the pressure shall be retained.

4.3.7 Preliminary run-in. The nature and extent of running-in shall be determined by the manufacturer, and shall be performed prior to the testing specified herein. All necessary adjustments other than normal control adjustments shall be made during this run-in and shall remain undisturbed thereafter.

4.4 Preproduction testing (see 6.2)

4.4.1 Preproduction test sample. One tank shall be subjected to the preproduction tests specified in 4.4.3.

4.4.2 Preproduction test report. After the preproduction tests are completed, three copies of a test report shall be prepared in accordance with MIL-STD-831 and submitted to the procuring activity.

* 4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the preproduction test report:

(a) A description of and the appropriate information below concerning any preproduction tests that were conducted prior to those described in the preproduction test report and that were intended to apply to the contract.

(1) All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

(2) Test operator and maintenance technicians errors, test equipment and test facility failures, and other events that might act as grounds for a request that an equipment failure not be counted as a reliability failure, shall be reported. Detailed descriptions of the events and the analysis to substantiate any such requests made shall be included and shall be clearly cross-referenced to each applicable failure.

(3) A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

* (4) A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made to production items or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections. Failures that have been corrected by design changes or by other means shall be counted as reliability failures until the corrections have been both analyzed and verified by test sufficient to substantiate the effectiveness of the correction to the satisfaction of the procuring activity.

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(5) Clock time and manhours required for each maintenance and servicing action taken during the tests. Only the time needed for actually preparing for and performing the tasks shall be measured and reported. A brief description of the experience and qualifications of the personnel taking such actions shall be included. The information shall include a summary of the data resulting from the servicing and maintenance tests required by 4.6.16. In addition, the clock time and the manhours required for actual access and disassembly shall be recorded for any teardown inspections done for mechanical checks in accordance with 4.6.17. The number and description of parts and assemblies removed and causing interference shall be noted for each teardown and inspection made. Administrative time, such as filling out records, and logistics time, such as obtaining parts from stock, shall not be included.

(6) Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

(7) Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

(b) Any of the above information that is already included in the preproduction test report body or in other documents submitted to the procuring activity need not be repeated in the information required by this paragraph, but clear reference to the location and to the date of submission of the data shall be included.

*(c) All table and column headings and all abbreviations and symbols used shall be clearly defined where used or in a table of abbreviation and symbol definitions placed in the front part of the appendix.

4.4.3 Preproduction test. The preproduction test shall consist of all the tests described under 4.6.

4.5 Acceptance test. The acceptance test shall consist of the following:

(a) Individual tests (4.5.1)

(b) Sampling plans and tests (4.5.2)

4.5.1 Individual tests

4.5.1.1 Spring-loaded relief valve test. Each spring-loaded relief valve shall be subjected to the test specified in 4.6.1.

4.5.1.2 Tank tests. Each tank shall be subjected to the following tests as described under 4.6:

(a) Examination of product (4.6.2)

(b) Functional check (4.6.3)

(c) Pressure test (4.6.4)

(d) Individual article vacuum retention testing (4.6.6.2)

(e) Evaporation loss rate individual acceptance test (4.6.7.2)

(f) Cleaning effectiveness tests (4.6.8)

4.5.2 Sampling plans and tests

4.5.2.1 Bursting disc sampling test

4.5.2.1.1 Bursting disc lot. A bursting disc lot shall consist of discs manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

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4.5.2.1.2 Bursting disc sampling and test. Three discs from each lot of fifty or less shall be selected at random and each subjected to a gradually increasing clean, dry, oil-free air or nitrogen pressure until it ruptures. In case a disc fails at a pressure outside the limits established by the tank manufacturer for compliance with this specification, five additional sample discs from the same lot shall be tested in the same manner.

4.5.2.1.2.1 Bursting disc rejection and retest. If more than one of the original samples or any of the five additional sample discs fail at a pressure outside the specified limits, the entire lot shall be rejected.

4.5.2.2 Valve sampling test

4.5.2.2.1 Valve lot. A lot shall consist of valves manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.2.2.2 Valve sampling and test. One of each type (same size, design, and construction) valve from each lot of one hundred or less shall be selected at random and subjected to the testing specified in 4.6.9. In case the valve fails to fulfill all specified requirements, three additional sample valves from the lot shall be tested in the same manner.

4.5.2.2.2.1 Valve rejection and retest. If any of the three additional sample valves fail to fulfill all specified test requirements, the entire lot shall be rejected.

4.5.2.3 Liquid line filter sampling test

4.5.2.3.1 Filter lot. A lot shall consist of filters manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.2.3.2 Filter sampling and test. One of each type (same size, design, and construction) filter from each lot of 75 or less shall be selected at random and subjected to the absolute and nominal rating tests specified in 4.6.15.1 and 4.6.15.2. In case a filter fails to fulfill all specified test requirements, two additional sample filters from the lot shall be tested in the same manner.

4.5.2.3.2.1 Filter reject and retest. If either of the two additional sample filters fails to fulfill all specified test requirements, the entire lot shall be rejected.

** 4.5.2.4 Dissimilar metal vacuum joint sampling tests

* 4.5.2.4.1 Joint lot. A lot shall consist of joints manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

* 4.5.2.4.2 Joint sampling and test. One of each size, style, and construction dissimilar metal vacuum joint from each lot of 25 or less shall be selected at random and subjected to the testing specified in 4.6.6.1.2. In case the joint fails to fulfill all specified requirements, three additional samples from the lot shall be tested in the same manner.

* 4.5.2.4.3 Joint rejection and retest. If any of the three additional sample joints fails to fulfill all specified test requirements, the entire lot shall be rejected.

4.6 Test methods

4.6.1 Spring-loaded relief valve test. Prior to installation on the tank, each spring-loaded relief valve shall be subjected to a gradually increasing pressure of clean, dry, oil-free air, oxygen,

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or nitrogen at its inlet until it starts to discharge. The pressure shall then be reduced until the valve completely reseals. This procedure shall be repeated until the valve has started to discharge, and resealed, not less than twice. Following this testing, the valve shall be checked for leakage by application of a soap film across the outlet and over all outside surfaces with not less than the maximum working pressure of the component which the valve will be used to protect, being applied at the valve inlet. Failure to open or reseal within the specified limits or an indication of leaking during the soap film test shall be cause for rejection.

4.6.2 Examination of product. The tank shall be inspected to determine whether it complies with the requirements specified herein with respect to dimensions, workmanship, finish, and marking. The tank shall be subjected to a thorough inspection to determine quality, ease of adjustment, alignment, and functioning of all parts.

4.6.3 Functional check. All mechanical parts shall be checked for free and proper functioning.

4.6.4 Pressure test. Each component and circuit of the tank that operates under positive pressure in normal service shall be pressure tested in accordance with section VIII of the ASME Boiler and Pressure Vessel Code. Automatic pressure relief devices need not be installed for this test. Any distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

4.6.5 Handling and transportability tests

*4.6.5.1 Forklift handling tests

*4.6.5.1.1 Single forklift test. A single forklift truck, with tines not more than 72 inches long and capable of handling the empty tank total weight, shall be used as follows to lift and move the empty tank:

(a) Raise the empty tank not less than 3 feet above the floor or ground, using the forklift entries adjacent to the tank center-of-gravity (see figure 3).

(b) Move the tank a horizontal distance of not less than 50 feet to a new location.

(c) Set the tank on the floor or ground at the new location, and

(d) Back the forklift truck not less than 10 feet away from the tank.

These lifting and movement procedures shall be accomplished not less than twice from each side of the tank. Any damage to or distortion of any tank component or interference with forklift handling shall be cause for rejection.

*4.6.5.1.2 Multiple forklift test. Two forklift trucks, with tines not more than 72 inches long and each capable of handling at least half the empty tank total weight, shall be used as follows to lift and move the empty tank:

(a) Raise the empty tank not less than 3 feet above the floor or ground, using the pairs of forklift entries nearest the ends of the skid base.

(b) Move the tank a horizontal distance of not less than 50 feet to a new location.

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(c) Set the tank on the floor or ground at the new location, and

(d) Back each forklift truck not less than 10 feet away from the tank. The lifting and movement procedures shall be performed as follows:

(1) Once with both forklift trucks on the same side of the tank.

(2) Repeat (1) above, except with both forklift trucks on the side of the tank opposite that used in (1).

(3) Once with one forklift truck on one side of the tank and the other forklift truck on the opposite side of the tank and at the opposite end.

(4) Repeat (3) above, except with each forklift truck repositioned on the same side of the tank as before but using the forklift entries at the opposite end of the skid base from those used in (3).

Any damage to or distortion of any tank component or interference with forklift handling shall be cause for rejection.

4.6.5.2 Hoisting provision test. Weights totaling not less than the empty weight of the completely assembled tank with all accessories shall be attached to the tank in such a manner that the combined weight of the tank and weights will be supported by the hoisting provisions specified in 3.5.7.3 when the tank is raised by an overhead crane. Filling an equivalent weight of liquid into the liquid storage tank assembly inner shell may be substituted for attachment of external weights if desired. The assembly (tank and weights) shall then be raised completely off the floor by an overhead crane to a height of not less than three feet, held in that position for not less than 1 minute, and then lowered to the floor, by means of the hoisting provisions. This procedure shall be repeated three times. Any evidence of damage to or distortion of any tank component shall be cause for rejection.

4.6.5.3 Air transport acceleration force testing. Capabilities of the tank design and construction fulfilling the air transport flight and taxiing acceleration forces and the emergency landing acceleration forces specified in MIL-A-8421 may be substantiated by calculations included as part of the test report specified in 4.4.2.

4.6.6 Vacuum retention tests

4.6.6.1 Preproduction article vacuum retention testing. Following the lifting provision testing specified in 4.6.5.1, the liquid storage tank assembly shall be filled with liquid oxygen or liquid nitrogen to not less than 60 percent of design capacity (unless already so filled) and permitted to temperature stabilized until a constant annular insulation space absolute pressure is obtained. After temperature stabilization, and with the inner shell vapor space at approximately atmospheric pressure, the annular insulation space absolute pressure shall be determined with an instrument of the type specified in 4.3.2.1.1. The inner shell vapor space shall then be pressurized to not less than 30 psig with oxygen or nitrogen gas obtained by vaporizing part of the contained liquid with the pressure buildup system, and the tank allowed to set for not less than 7 days (168 hours) so filled and pressurized. The pressure shall be controlled by use of the inner shell vapor space pressure control specified in 3.8.1.9.3 and the pressure buildup system specified in 3.8.2. Freeze-up of the pressure control device or interference with venting shall be cause for rejection.

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4.6.6.1.1 Following completion of the test period specified in 4.6.6.1, the liquid storage tank assembly inner shell vapor phase pressure shall be reduced to atmospheric and the annular insulation space absolute pressure determined with an instrument of the type specified in 4.3.2.1.1. The annular insulation space absolute pressure shall not exceed 100 microns Hg, or show any increase attributable to leakage or outgassing, or the tank shall be considered to have failed the test.

* 4.6.6.1.2 Dissimilar metal vacuum joints used in the tank construction shall be tested as follows:

* (a) Subject the joint assembly to 10 thermal shocks. Each thermal shock shall be accomplished by first heating the complete joint assembly to not less than 125°F and then, before the joint has cooled below 125°F, quickly submerging the complete joint assembly in liquid nitrogen at its atmospheric pressure boiling temperature. The joint assembly shall remain submerged in the liquid nitrogen until the joint has cooled to approximately liquid nitrogen temperature. The joint assembly will be considered to have reached liquid nitrogen temperature when bubbles cease to emanate from its surface in large numbers.

* (b) Secure the joint assembly by one end only (with the other end free to move) to a vibration input device which will vibrate the joint assembly in liquid nitrogen. The joint assembly shall then be subjected to axial vibration at a frequency of 60 cps and a double amplitude of not less than 0.050 inch for not less than 30 minutes while the joint assembly is submerged in liquid nitrogen at its atmospheric pressure boiling temperature.

* (c) Secure the joint assembly as in (b), above. The joint assembly shall then be subjected to lateral vibration at a frequency of 60 cps and a double amplitude of not less than 0.200 inch for not less than 30 minutes while the joint is submerged in liquid nitrogen at its atmospheric pressure boiling temperature.

* Any leakage rate increase detectable with a helium mass spectrometer operating at its maximum sensitivity, comparing checks prior to and following the temperature shock and vibration testing specified above, shall be cause for rejection.

4.6.6.2 Individual article vacuum retention testing. The liquid storage tank assembly inner shell shall be filled with liquid oxygen or liquid nitrogen to not less than 60 percent of design capacity and permitted to temperature stabilize. Following temperature stabilization of the liquid storage tank assembly (as indicated by a constant annular insulation space absolute pressure), and with the inner shell vapor space at approximately atmospheric pressure, the annular insulation space absolute pressure shall be determined and the tank tested for leakage or outgassing as specified in 4.6.6.1 and 4.6.6.1.1. The annular insulation space absolute pressure shall not exceed 100 microns Hg following the test or show any increase attributable to leakage or outgassing, or the tank shall be considered to have failed the test.

4.6.7 Heat leak evaporation loss rate tests

4.6.7.1 Initial preproduction test. The initial preproduction heat leak evaporation loss rate test shall consist of the following individual test operations conducted in the order listed. Throughout the complete test, the inner shell vapor phase pressure shall remain at as near

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atmospheric as necessary pressure drops through the measuring apparatus will permit.

4.6.7.1.1 Following the vacuum retention test and with the liquid storage tank assembly inner shell remaining filled to not less than 50 percent of design capacity with liquid oxygen or liquid nitrogen, the tank shall be placed in a location where the ambient temperature will remain at not less than 125°F throughout the periods specified.

4.6.7.1.2 The inner shell vapor space shall be vented to atmospheric pressure for not less than 24 hours through the flow measuring apparatus that will be used for determining the evaporation loss rate.

* 4.6.7.1.3 The tank shall remain under the specified ambient temperature and inner shell vapor phase pressure conditions for an additional 72 hours while heat leak evaporation rate losses for the tank are determined by passing all vapors vented from the inner shell through a recording- or totaling-type gas flowmeter. During this 72-hour period, the upper surface of the tank shall be exposed to the full impact of solar or equivalent heat radiation of 100 to 120 watts per square foot for three separate periods of not less than 8 hours each, with each exposure followed by a period of not less than 16 hours during which no radiation is applied. The total weight of vapors vented during the test, the weight of vapors vented since the preceding reading, the average ambient temperature to which the tank has been subjected during the preceding period, and the average barometric pressure to which the tank has been subjected during the preceding period shall be recorded at intervals of not to exceed 8 hours during the 72-hour test. The actual loss rate for each reading, the average actual loss rate over the entire 72-hour test period, and the average ambient temperature to which the tank was exposed during the 72-hour test period shall be calculated. The evaporation loss rates shall be reported as the pounds of liquid oxygen vaporized per 24-hour day, and shall not exceed the limit specified in 3.7.2, or the tank shall be considered to have failed the test.

4.6.7.1.4 The tank shall be removed from the high temperature test conditions and exposed to ambient temperature conditions for not less than 36 hours to permit temperature stabilization.

4.6.7.1.5 The tank shall be subjected to the individual acceptance evaporation loss rate test specified in 4.6.7.2 to establish comparative data for determining whether production tanks are capable of complying with the requirements of 3.7.2.

* 4.6.7.2 Evaporation loss rate individual acceptance test. Following the vacuum retention test and with the inner shell remaining filled to not less than 50 percent of design capacity, the tank shall be placed where it will be subjected to local or shop ambient temperatures throughout the test. The inner shell vapor space shall be vented to atmospheric pressure for not less than 24 hours through the flow measuring apparatus that will be used for determining the evaporation loss rate. The heat leak evaporation loss rate for the tank shall then be determined by passing all vapors vented from the inner shell through a recording- or totaling-type gas flowmeter for an additional period of not less than 72 hours. Throughout the test, the inner shell vapor phase pressure shall remain at as near atmospheric as necessary pressure drops through the flow measuring apparatus will permit.

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* 4.6.7.2.1 The total weight of vapors vented during the test, the weight of vapors vented since the preceding reading, the average ambient temperature to which the tank has been subjected during the preceding period, and the average barometric pressure to which the tank has been subjected during the preceding period shall be recorded at intervals of not to exceed 8 hours throughout the 72-hour test. The actual loss rate in pounds for each reading, the average actual loss rate in pounds over the entire 72-hour test period, and the average ambient temperature to which the tank was exposed during the 72-hour test period shall be calculated. The production item average loss rate for the 72-hour test period shall be corrected from the average temperature experienced during the test period to the average temperature at which the comparative data specified in 4.6.7.1.5 was obtained, and the result reported as the pounds of liquid oxygen vaporized per 24-hour day. This loss rate shall not exceed the limit established by the comparative data obtained in 4.6.7.1.5 for compliance with 3.7.2 or the tank shall be considered to have failed the test.

4.6.8 Cleaning effectiveness tests

4.6.8.1 Visual inspection. All equipment, pipes, and components that will contact liquid oxygen shall be visually examined for evidence of corrosion products, metal chips, scale, weld scale, oil, grease, paints, preservatives, decals, or other contamination of foreign matter. Any evidence of contamination or foreign matter shall require recleaning and retest.

4.6.8.2 Ultraviolet light inspection. All accessible surfaces and parts that will contact liquid oxygen shall be visually inspected for hydrocarbons with ultraviolet light. Any evidence of fluorescence shall require recleaning and retest.

4.6.8.3 Particulate matter test. Following the cleaning and degreasing operations specified herein and equipment final assembly, the liquid storage tank inner shell shall be filled to not less than 90 percent of design capacity with liquid oxygen or liquid nitrogen. The tank shall then be permitted to set for at least 2 hours undisturbed. Following the 2-hour period, a 2-liter or larger Dewar flask shall be placed under the discharge end of the fill-drain line specified in 3.8.1.8.1.1.1 with its filter removed, the fill-drain line shutoff valve opened, and not less than 1 liter of the first liquid discharged through the line collected in the Dewar flask. This sample shall be evaporated to dryness and the residue examined for particulate matter. The liquid storage tank and the associated piping and fittings shall be recleaned and this test repeated until the particulate matter remaining as residue contains:

(a) No solid particle with any dimension greater than 1,000 microns

(b) No fibrous particles with a length greater than 6,000 microns

(c) Not more than 25 milligrams total of both solid and fibrous particles.

4.6.8.4 Total solids test. Following withdrawal of the 1-liter sample specified in 4.6.8.3, the entire load of contained liquid shall be discharged at a flow rate of not less than 90 gpm through the fill-drain line specified in 3.8.1.8.1.1.1 and the filter specified in 3.8.1.8.1.4. The filter shall be thoroughly cleaned and the removed material collected, dried, and weighed after each use. If more than 0.25 gram of contaminant material is collected on the filter during discharge of the liquid from the tank, the liquid storage tank assembly and the associated piping and

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fittings shall be recleaned. Cleaning and testing shall be repeated until the total amount of contaminant collected on the filter from discharging the liquid storage tank assembly load is less than 0.25 gram.

4.6.9 Shutoff and control valve tests. The valve shall be subjected to the following tests in the order specified:

4.6.9.1 Cycling test. The valve shall be closed with a torque of 60 ±5 pound inches per inch of nominal size, and subjected to a compressed air, oxygen gas, or nitrogen gas inlet pressure of not less than 50 psig with the outlet open to atmospheric pressure. The valve shall then be opened to not less than the three-quarters open position, after which it shall be again closed with a torque of 60 ±5 pound inches per inch of nominal size and the specified inlet pressure re-established. This shall constitute one cycle. One thousand such cycles shall be accomplished without intervening valve lubrication, adjustment, or repair while the valve body is subjected to ambient temperature. The thousand cycle test shall then be repeated with the valve body submerged in liquid oxygen or liquid nitrogen. Damage to or improper operation of the valve as a result of the testing shall be cause for rejection.

4.6.9.2 Overtorque test. At the conclusion of the cycling testing specified in 4.6.9.1, the valve shall be removed from the liquid bath and permitted to stabilize at ambient temperature. The valve shall then, without intervening lubrication, adjustment, or repair, be opened fully, and closed with a torque of 300 ±10 pound inches per inch of nominal size. Damage to or improper operation of the valve as a result of the testing shall be cause for rejection.

4.6.9.3 Leakage test. Following the overtorque testing specified in 4.6.9.2 and without intervening valve lubrication, adjustment, or repair, the valve shall be fully opened, and closed with a torque of 60 ±5 pound inches per inch of nominal size. The valve shall then be completely immersed in clear water with the inlet subjected to a compressed air, oxygen gas, or nitrogen gas pressure of not less than 50 psig for not less than 10 minutes with the outlet open. Any leakage past the seat, through the packing, or through the valve body shall be collected and measured. Total collected gas leakage exceeding 2 cubic inches of free air, nitrogen gas or oxygen gas per hour per inch of valve nominal size shall be cause for rejection.

* 4.6.10 Disconnect coupling torque resistance tests. The liquid fill-drain line disconnect coupling and the servicing line disconnect fitting shall each be subjected to a torque of not less than 300 lb ft per inch of nominal line size applied to the coupling or fitting first in one and then in the other direction of rotation with a 24-inch wrench. Any twisting of, distortion of, or damage to tank piping or connections resulting from this torque testing shall be cause for rejection.

4.6.11 Dimensions and weight. The overall dimensions and total weight, including all equipment and accessories, of the completely assembled, empty tank shall be reported on the test data sheets. The dimensions and weight shall not exceed the limits specified in 3.10 and 3.11.

4.6.12 Liquid storage capacity test. The tank storage capacity shall be determined by filling a liquid into the inner shell with the tank setting on its skid base on a flat, level surface until the filling liquid overflows through the full trycock line specified

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in 3.8.1.8.1.9, recording the liquid level gage reading in terms of liquid oxygen, and then either measuring the volume of the filling liquid or determining its volume from weights before and after filling. Following this, the inner shell shall be filled until the filling liquid overflows through the vapor vent line specified in 3.8.1.8.1.7, and the quantity of filling liquid determined as above. The inner shell shall then be filled completely to determine the total internal volume, and the quantity of filling liquid again determined as above. The actual liquid volume when the inner shell is filled to overflow through the full trycock line, when filled to overflow through the vapor vent line, and when completely filled shall all three be recorded in U.S. gallons and shall be accurate to within ±5 gallons. The tank storage capacity at which liquid starts to discharge through the full trycock line, the capacity at which liquid starts to discharge through the vapor vent line, and the total internal volume shall comply with 3.7.1, 3.8.1.8.1.7, and 3.8.1.8.1.9, or the tank shall be considered to have failed the test.

4.6.13 Pressure buildup tests

4.6.13.1 Buildup with low liquid level. The liquid storage tank assembly shall be filled with enough liquid oxygen or liquid nitrogen to insure between 100 and 125 gallons remaining in the inner shell after the specified stabilization. The tank shall then be set with the inner shell vapor space vented to atmospheric pressure for not less than 4 hours. Following this stabilization period, the inner shell vapor vent line valve shall be closed and the pressure buildup valve opened. The inner shell vapor phase pressure shall reach 30 psi or the tank shall be considered to have failed the test.

4.6.13.2 Pressure buildup rate test. The liquid storage tank assembly shall be filled with liquid oxygen or liquid nitrogen to between 15 and 20 percent of capacity and permitted to set with the inner shell vapor space vented to atmospheric pressure for not less than 4 hours and until all external components of the tank reach approximately ambient temperature. The inner shell vapor vent line shutoff valve shall then be closed and the pressure buildup system actuated. The length of time required after closing of the vent valve until the vapor phase pressure has reached 30 psig shall be recorded on the test data sheets and shall not exceed 10 minutes, or the tank shall be considered to have failed the test.

4.6.14 Liquid transfer tests

4.6.14.1 Fill-drain line transfer test. Liquid oxygen shall be transferred into the tank through the fill-drain line specified in 3.8.1.8.1.1.1 at not less than 100 gpm for a length of time adequate to determine the pressure drop resulting from friction losses in the filter and line. The pressure drop shall not exceed that permitted in 3.8.1.8.1.1.1, or the tank shall be considered to have failed the test.

4.6.14.1.1 Liquid nitrogen may be used at a flow rate which will provide a pressure drop equivalent to that resulting from the specified liquid oxygen flow. If liquid nitrogen is employed for this purpose, data shall be provided to substantiate the flow rate used.

4.6.14.2 Servicing line transfer test. Liquid oxygen shall be transferred from the tank through the servicing line specified in 3.8.1.8.1.5 at not less than 25 gpm for a length of time adequate to determine the pressure drop resulting

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from friction losses in the filter and line. The pressure drop shall not exceed that permitted in 3.8.1.8.1.5, or the tank shall be considered to have failed the test.

4.6.14.2.1 Liquid nitrogen may be used at a flow rate which will provide a pressure drop equivalent to that resulting from the specified liquid oxygen flow. If liquid nitrogen is employed for this purpose, data shall be provided to substantiate the flow rate used.

4.6.14.3 Vapor vent line test. The tank shall be subjected to conditions that will insure all liquid storage tank assembly components attaining a temperature of not less than 125°F. After the liquid storage tank assembly components have stabilized at this temperature, liquid oxygen or liquid nitrogen shall be transferred into the inner shell through the fill-drain line until the inner shell has cooled sufficiently to permit filling at the 100 gpm rate specified in 3.8.1.8.1.7. The rate of transfer during cooldown shall be the maximum that will maintain a liquid storage tank assembly inner shell vapor phase pressure of less than 15 psig. The cooldown time shall not exceed 15 minutes, or the tank shall be considered to have failed the test. Following cooldown, the liquid storage tank assembly shall be filled at a rate of not less than 100 gpm. The inner shell vapor phase pressure shall not exceed that specified in 3.8.1.8.1.7 when filled at the specified rate, or the tank shall be considered to have failed the test.

4.6.14.4 Vapor relief capacity test. Following the vapor vent line test specified in 4.6.14.3 and with the tank remaining filled to not less than 90 percent of design capacity, the vapor vent line shutoff valve specified in 3.8.1.8.1.7.1 shall be closed tightly, the inner shell vapor space pressure control specified in 3.8.1.9.3 rendered inoperative, the pressure buildup system set for maximum pressure buildup rate, and all other valves closed. The tank excess pressure shall be permitted to vent through the relief valve specified in 3.8.1.9.3.1.1 until the vapor phase pressure reaches its maximum reading. The vapor phase pressure shall not exceed 65 psig under these conditions, or the tank shall be considered to have failed the test.

4.6.15 Filter tests

4.6.15.1 Absolute rating. Compliance with the 40-micron absolute rating requirements of 3.8.1.8.1.4.2 shall be substantiated by the bubble point method as follows: The filter element shall be submerged in Solox 190, or equivalent, fluid. The fluid level shall be maintained at not to exceed 1/2 inch above the top of the element. The element shall be slowly pressurized from within and slowly rotated 360 degrees at each pressure increment. If the first bubble is emitted from the element at a pressure of less than 6 inches of water, the filter shall be considered to have failed the 40-micron requirement. The pressure shall be increased until the bubble point is determined.

4.6.15.2 Nominal rating. Compliance with the removal of 98 percent by weight of all particles whose smallest dimension is 10 microns or greater as required by 3.8.1.8.1.4.2 shall be substantiated as follows: Approximately 10 liters of liquid oxygen or liquid nitrogen previously

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filtered through a 5-micron millipore, or equivalent, membrane filter shall be contaminated with 10 grams of particles with the following size distribution:

Size Of Particle (Microns)	Percentage By Weight
10 to 20	36 ±3
20 to 40	24 ±3
40 to 60	16 ±3
Over 60	24 ±3

4.6.15.2.1 The liquid shall be placed in a container, agitated to insure a homogenous mixture, and then transferred through the filter in a time interval of not more than 2 minutes. The filter shall be so positioned that the flow is vertically down and the discharge is directed into a clean particle-free container with provisions to insure positive retention of any particles passing through the filter and exclusion of external particles. The liquid shall be permitted to vaporize and the container flushed by use of a wash bottle containing approximately 200 milliliter (ml) of isopropyl alcohol which has been previously filtered through a 5-micron millipore, or equivalent, membrane filter. The wash solutions shall then be filtered through a preweighed 5-micron millipore, or equivalent, filter. This container wash procedure shall be performed four times. The filter funnel shall then be washed with an additional 200 ml of solvent. Upon completion of the filtering, the millipore filter shall be weighed again. If the weight increase is more than 0.2 gram, the filter shall be considered to have failed the 98 percent retention of all particles 10 microns or larger.

4.6.15.3 Liquid fill-drain line filter pressure drop test. Liquid oxygen shall be transferred through the filter specified in 3.8.1.8.1.4 at not less than 100 gpm for a length of time adequate to determine the pressure drop through the filter. The liquid oxygen stream shall be contaminated with 5 grams of particles of the size and composition listed in table I. The 5 grams of contaminants shall be introduced into the liquid oxygen stream at the approximate rate of 1 gram per minute. Introduction of the contaminant shall not require more than 4-1/2 minutes ±30 seconds. The pressure drop, after the filter has been contaminated, shall not exceed 3 psi, or the filter shall be considered to have failed the test. The filter shall then be further contaminated with particles until a constant pressure drop of not less than 50 psig across the filter element is attained. Following this, the filter shall be opened and inspected. Any permanent distortion or damage incurred by the filter element or housing as a result of the 50 psig pressure differential shall be cause for rejection.

*4.6.15.3.1 Alternate test material. Water may be used at a flow rate which will provide a pressure drop equivalent to that resulting from the specified liquid oxygen flow. If water is employed for this purpose, data shall be provided to substantiate the flow rate used.

4.6.15.4 Servicing line filter pressure drop test. The servicing line filter specified in 3.8.1.8.1.5.4 shall be tested as specified in 4.6.15.3, except that the flow rate shall be not less than 25 gpm.

4.6.16 Wind test. It shall be demonstrated, by testing or calculation, that the tank will withstand the 70-mph wind specified in 3.8.3.3 and 3.8.3.3.1.

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4.6.17 Servicing and maintenance test. All normal preventive maintenance and servicing operations specified in the maintenance and instruction handbook prepared by the contractor shall be performed to determine their adequacy, ease of accomplishment, and the accessibility of parts and assemblies for performance of same, unless such instructions are contrary to those necessary for compliance with the requirements specified herein. Insofar as practicable, these operations shall be conducted as part of the normal preventive maintenance, servicing, and inspections performed in accomplishing the testing specified herein. The servicing shall include vacuum pumping and sealing of the vacuum line. Interference or obstructions to servicing or preventive maintenance shall be reported in detail on the test data sheets. The clock time and the man-hours required for each task shall be measured.

4.6.18 Mechanical check. Upon completion of the above tests, a critical inspection shall be made of components to determine their operability and any damage or undue wear incurred during the tests. Teardown and parts measurement shall be made only in those cases where service life is in question. Where teardown and parts measurement are performed, wear or distortion that exceeds limits permitted by the manufacturer for new parts shall be cause for considering the part or parts affected as having failed to complete the test satisfactorily. The clock time and manhours required for actual access, removal, and disassembly shall be measured for all teardowns made.

4.6.18.1 Steel parts subject to high stress in operation and that are suspected of having defects shall also be subjected to magnetic particle inspection (magnaflux) and shall exhibit no indications of damage attributable to the tests, or shall be considered as having failed the test. Grinding checks and subsurface indications of laps or seams shall not be cause for rejection. Magnetic particle inspection shall be performed in accordance with MIL-I-6868.

4.6.18.2 Nonmagnetic parts suspected of defects shall be subjected to inspection with fluorescent penetrant (black light) and shall exhibit no breaks or other defects that would impair their life or usefulness. Penetrant inspection shall be performed in accordance with MIL-I-6866.

4.6.19 Servicing cycle test. The liquid storage tank assembly shall be filled to design capacity with liquid oxygen or liquid nitrogen and permitted to set with the inner shell vapor space vented to atmospheric pressure until all components, other than the vapor vent line, external to the liquid storage tank assembly reach approximately ambient temperatures. The tank shall then be subjected to the testing specified in 4.6.19.1 through 4.6.19.3.

4.6.19.1 Following the filling and stabilization specified in 4.6.19, a servicing cycle shall be performed. The servicing cycle operation shall be accomplished as follows:

(a) Connect the tank transfer hose disconnect fitting to a container suitable for use with liquid oxygen or liquid nitrogen.

(b) Close the vapor vent line shutoff valve.

(c) Open the pressure buildup valve until the inner shell vapor phase pressure has risen to 30 psig, after which close the pressure buildup valve.

(d) Fully open the servicing line shutoff valve and transfer not less than 10 gallons of liquid through the servicing hose into the receiving container.

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(e) Close the servicing line shutoff valve.

(f) Open the vapor vent line shutoff valve to vent the inner shell to atmospheric pressure.

(g) Permit the tank to set with the vapor vent line shutoff valve fully open until all traces of ice and frost have disappeared from the pressure buildup coil, tank valves, and tank servicing line. The application of external heat to expedite warmup is permissible.

4.6.19.2 The servicing cycle specified in 4.6.19.1 shall be repeated until the storage tank has been subjected to not less than 50 such cycles.

4.6.19.3 The tank shall complete not less than 50 servicing cycles without necessity for servicing or maintenance of any kind, or shall be considered to have failed the test.

4.6.20 Reliability demonstration and test. Satisfactory completion of all tests required herein will be considered to demonstrate acceptable compliance with the quantitative reliability requirements of this specification.

4.6.21 Maintainability evaluation. The tank shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Examination of preparation for delivery. Preservation, packaging, packing, and marking shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 General instructions

5.1.1 Cleaning. The tank shall be thoroughly cleaned to remove excess and spilled lubrication materials, loose or chipped paint, spilled chemicals, and other foreign materials. All cleaning solvents shall be thoroughly removed from the tank components and accessories prior to delivery.

5.1.2 Repainting. Tank surfaces on which the paint is damaged or defective shall be thoroughly cleaned and repainted as specified under 3.12.

5.1.3 Liquid storage tank assembly purging and sealing. The liquid storage tank assembly inner shell shall be drained of all liquids and purged with not less than 500 standard (68°F and 760 mm Hg) cubic feet of clean, dry, oil-free nitrogen gas introduced through the vapor phase vent line specified in 3.8.1.8.1.7. The liquid fill-drain and servicing line shutoff valves shall be opened individually and the gas flow permitted to purge the respective lines, valves, and fittings. The bulk of the purge gas shall be exhausted through the liquid fill-drain line specified in 3.8.1.8.1.1.1.

5.1.3.1 Following the purging, the inner shell shall be pressurized to not less than 10 psig nor more than 20 psig with clean, dry, oil-free nitrogen gas, after which every valve and line leading into the inner shell shall be closed. The ends of all pipes, tubes, and couplings shall be securely sealed with pressure-sensitive tape conforming to PPP-T-60. The liquid storage tank assembly annular insulation space

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shall be evacuated to the absolute pressure recommended in the tank operating and servicing instruction handbook if it is not at or below that pressure. The vacuum line shutoff valve shall be tightly closed after the evacuation and the vacuum line sealed with the cap provided.

5.1.3.2 A Type A tag conforming to UU-T-81, with tag and printing waterproofed, stating that the liquid storage tank assembly inner shell is cleaned, and pressurized with clean, dry, oil-free nitrogen gas, and indicating the date and the pressure to which the annular insulation space has been evacuated, shall be securely attached to the vapor vent line shutoff valve handle in a conspicuous location.

5.1.4 Preservatives. Preservatives and lubricants shall not be applied to any part of the equipment that will come into contact with high-purity oxygen.

5.1.5 Securing for shipment. Small items and packages not fastened to major components shall be secured to or packed with major components to insure against accidental loss or damage caused by movement during shipment.

5.2 Physical protection. Cushioning, blocking, bracing, bolting, and testing as necessary to insure compliance with the requirements specified herein shall be in accordance with MIL-STD-1186.

5.3 Preservation and packaging for overseas or domestic shipment (level A, B, or C).

5.3.1 Preservation. Preservation shall consist of the procedures specified in 5.1.

5.4 Exterior packing. Unless otherwise specified (see 6.2), each tank shall be packed for storage and shipment in accordance with the following:

5.4.1 Domestic or overseas (level A, B, or C). In addition to the requirements of 5.3.1, each tank shall be provided with a protective shield constructed of wood for the purpose of protecting the instruments and control fittings. The wooden protective shield shall enclose the entire instrument and control area and shall be securely fastened to the tank. The shield shall be suitably constructed to provide protection to the instruments and controls while the tank is in transit or storage. Exterior shipping containers are not required.

5.5 Marking of shipments. Marking shall be in accordance with MIL-STD-129. The nomenclature shall be as follows:

TANK, STORAGE, LIQUID OXYGEN,
2000 GALLON CAPACITY, TYPE
TMU-7A/E, SKID MOUNTED

6. NOTES

6.1 Intended use. The TMU-7A/E tank is intended for storage at air bases of liquid oxygen with low evaporation loss rates until needed. The stored liquid oxygen is then transferred from this tank to trailer-mounted tanks for distribution to aircraft systems requiring liquid oxygen.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether a complete set of all special tools, other than common handtools and

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those normally available in motor vehicle repair stops, are to be provided (see 3.8.6).

(c) Location and conditions for preproduction testing (see 4.4).

(d) That, when specified, the preproduction tests may be waived if the contractor's item has previously passed the preproduction tests specified herein.

(e) Required level of preservation, packaging, and packing (see section 5).

6.3 Definitions. For the purpose of this specification, the following definitions will apply.

6.3.1 MTBF. MTBF is the average (arithmetic mean) of the operating time between failures or the mean-time-between-failure. Note that the point estimate of MTBF from demonstration data (the sum of the test operating time divided by the number of counted failures) is larger than the MTBFmin, demonstrated at a specified confidence as described below. Note also, that the MTBFmin, considered acceptable and to be demonstrated is often lower than the MTBF reasonably expected and attainable within the current state-of-the-art. MTBF is often used as an expression representing MCBF, mean-cycles-between-failure.

6.3.2 Servicing cycle. A servicing cycle is defined as pressurizing the liquid storage tank assembly inner shell vapor space, transferring liquid into another vessel, and depressurizing the inner shell vapor space back to approximately atmospheric pressure.

6.3.3 Ambient conditions. Ambient conditions are the dry-bulb temperature, wet-bulb temperature, and the relative humidity of the atmospheric air surrounding and in the vicinity of the tank, but unaffected by any heat or cold emanating from the tank itself.

6.3.4 Dissimilar metal vacuum joint. A dissimilar metal vacuum joint is a mechanical (welded or other type) joint between different base metals, or between alloys with different base metals as the principal constituent, used in the tank construction where the joint must function to seal the liquid storage tank assembly annular insulation space from a different pressure.

6.3.5 Pressures. All pressures referred to herein, unless specified as absolute or denoted by the symbol psia, will be interpreted as pounds per square inch (psi) gage.

6.3.6 Valve cycle of operation. A cycle of operation for a valve is defined as opening the valve from sealing a differential pressure of not less than 50 psig to the not less than three-fourths open position, and reclosing the valve with a torque of 60 ±5 pound-inches per inch of valve nominal size.

6.3.7 Fibrous particle. A fibrous particle is defined as a long, slender particle whose maximum cross-sectional dimension is 40 microns.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors

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are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 11
Navy - AS

Preparing activity:
Air Force - 11

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Navy - AS, MC

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-T-27292F(ASG) Tank, Storage, Liquid Oxygen TMU-7A/E

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

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